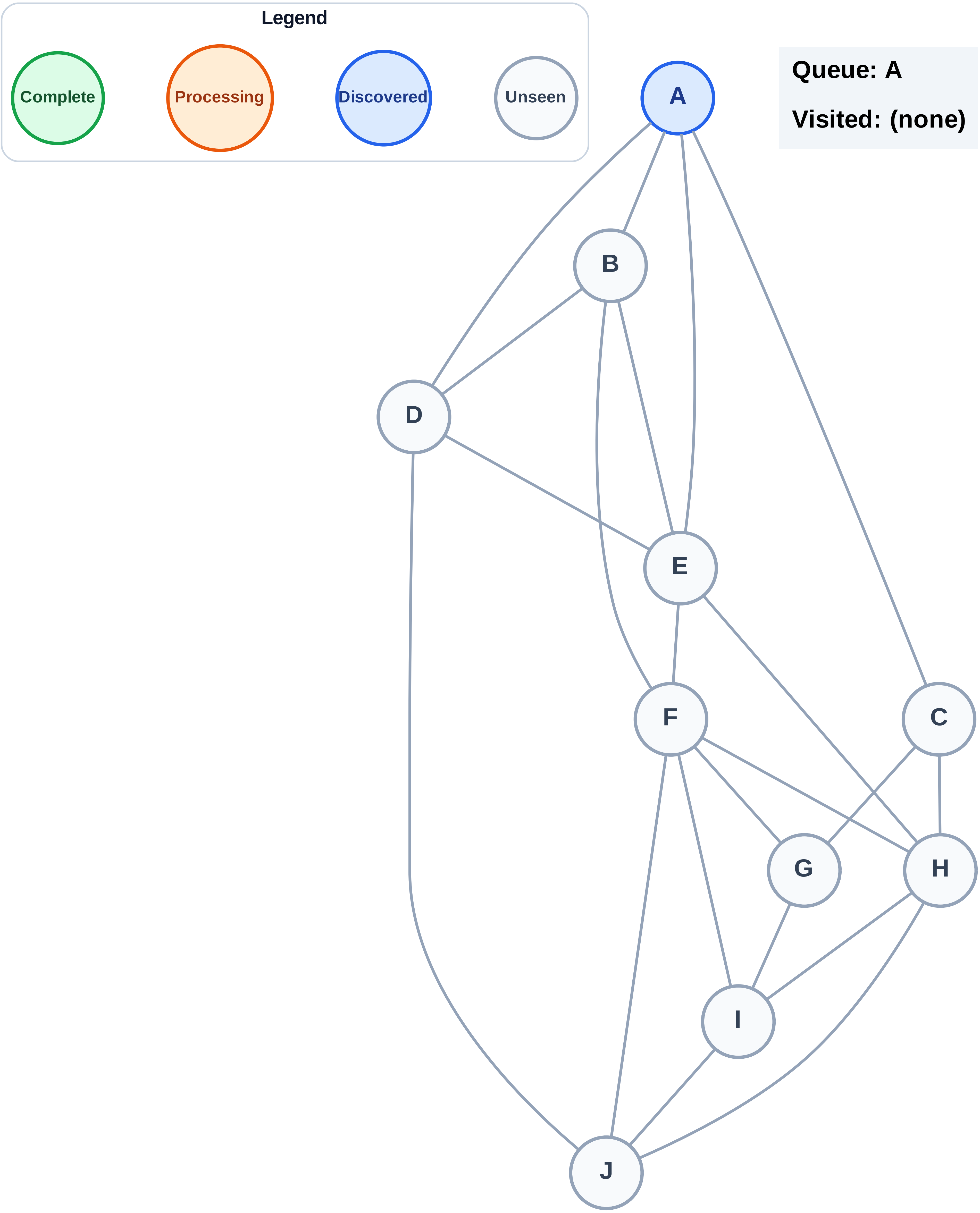
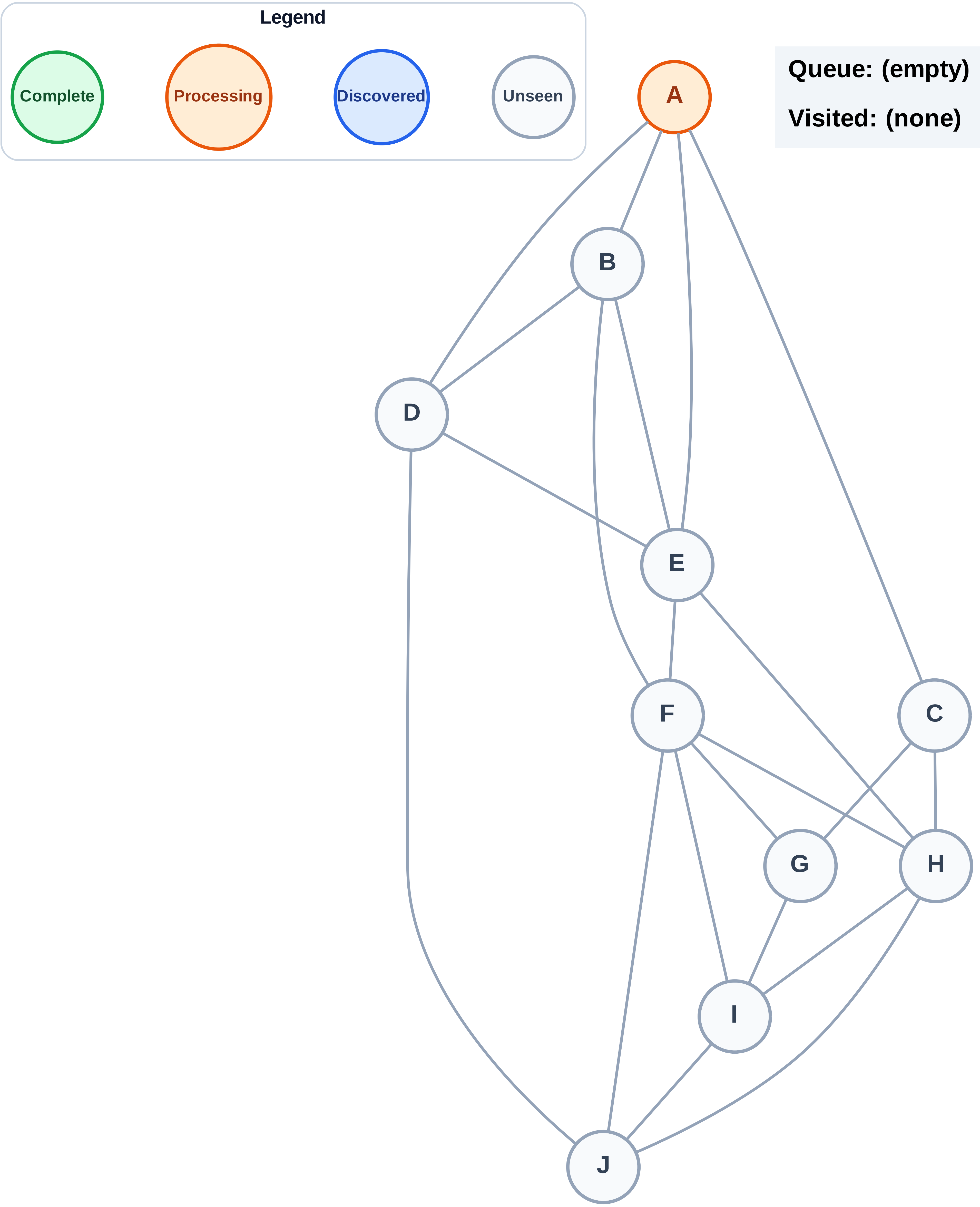
BFS Traversal

Step 1: Start at A



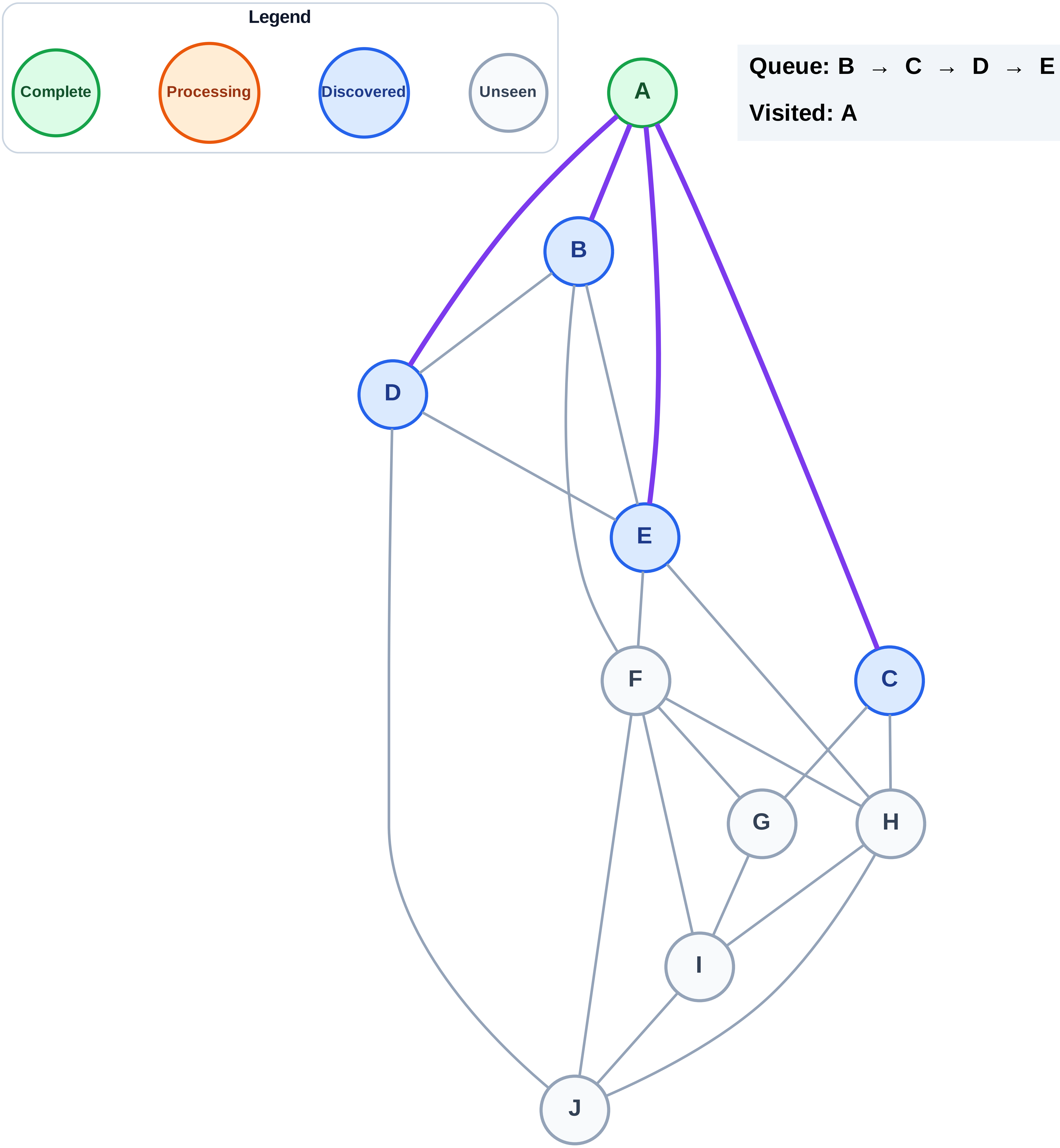
BFS Traversal

Step 2: Process node A



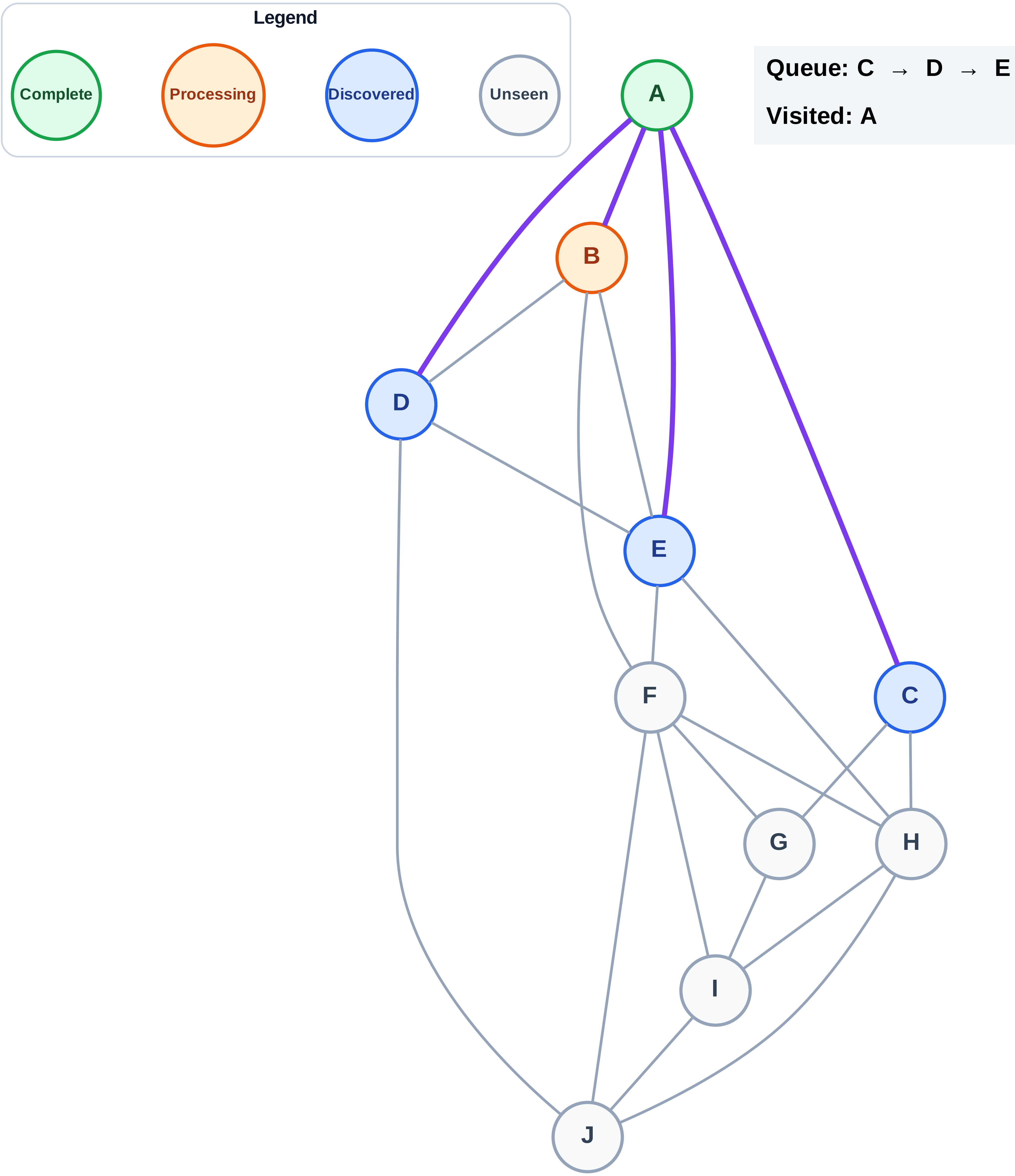
BFS Traversal

Step 3: Discover B, C, D, E from A



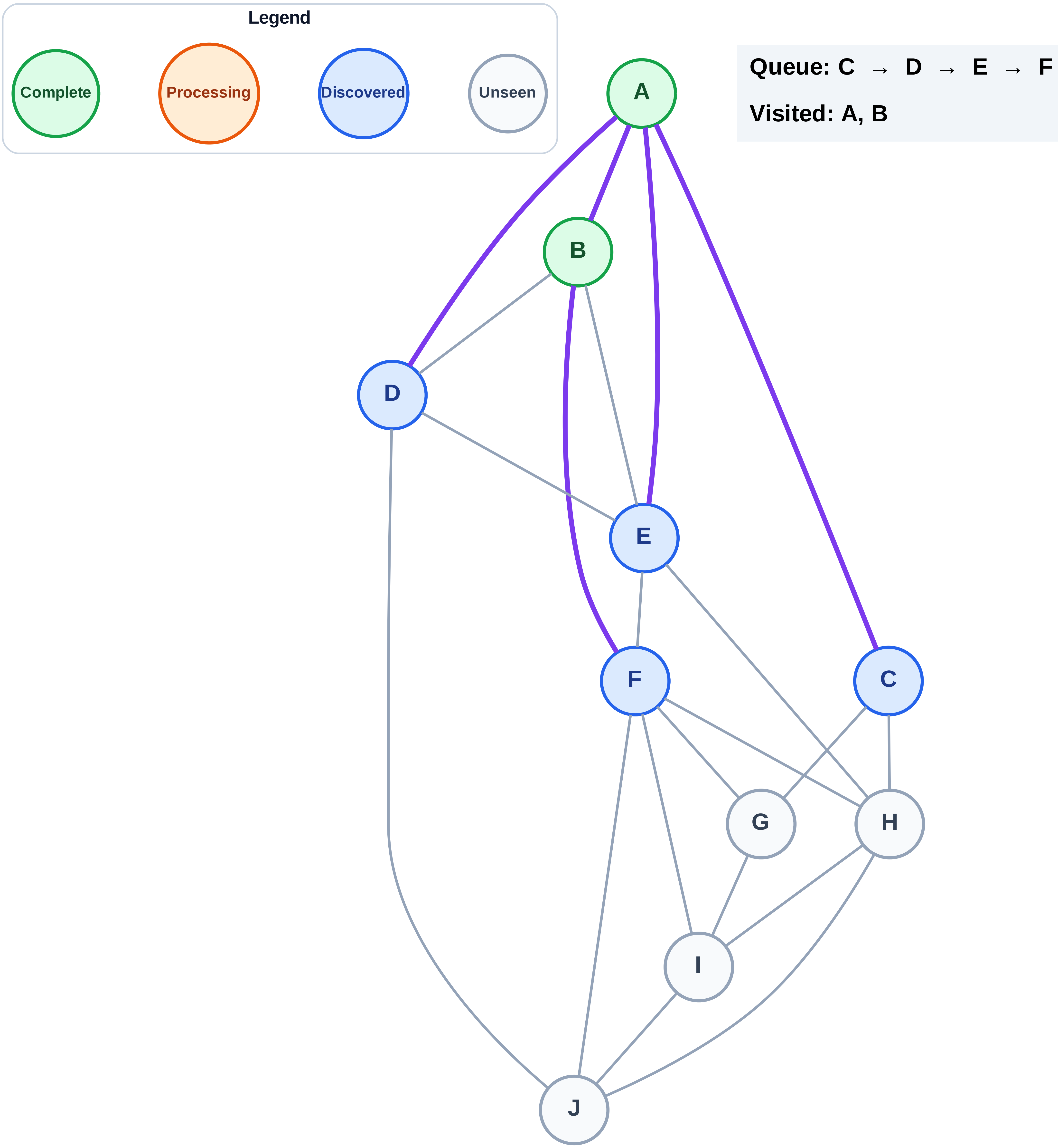
BFS Traversal

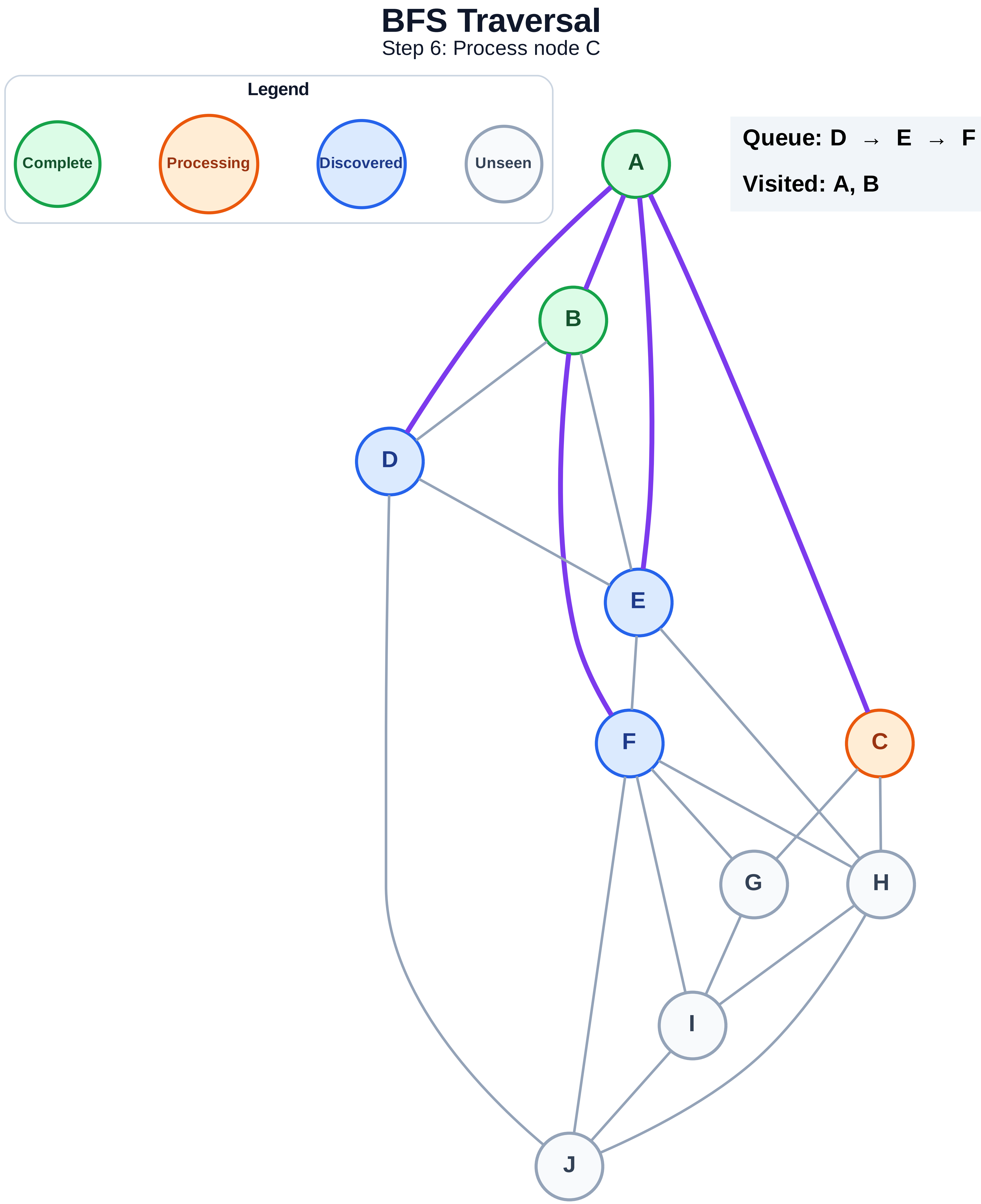
Step 4: Process node B



BFS Traversal

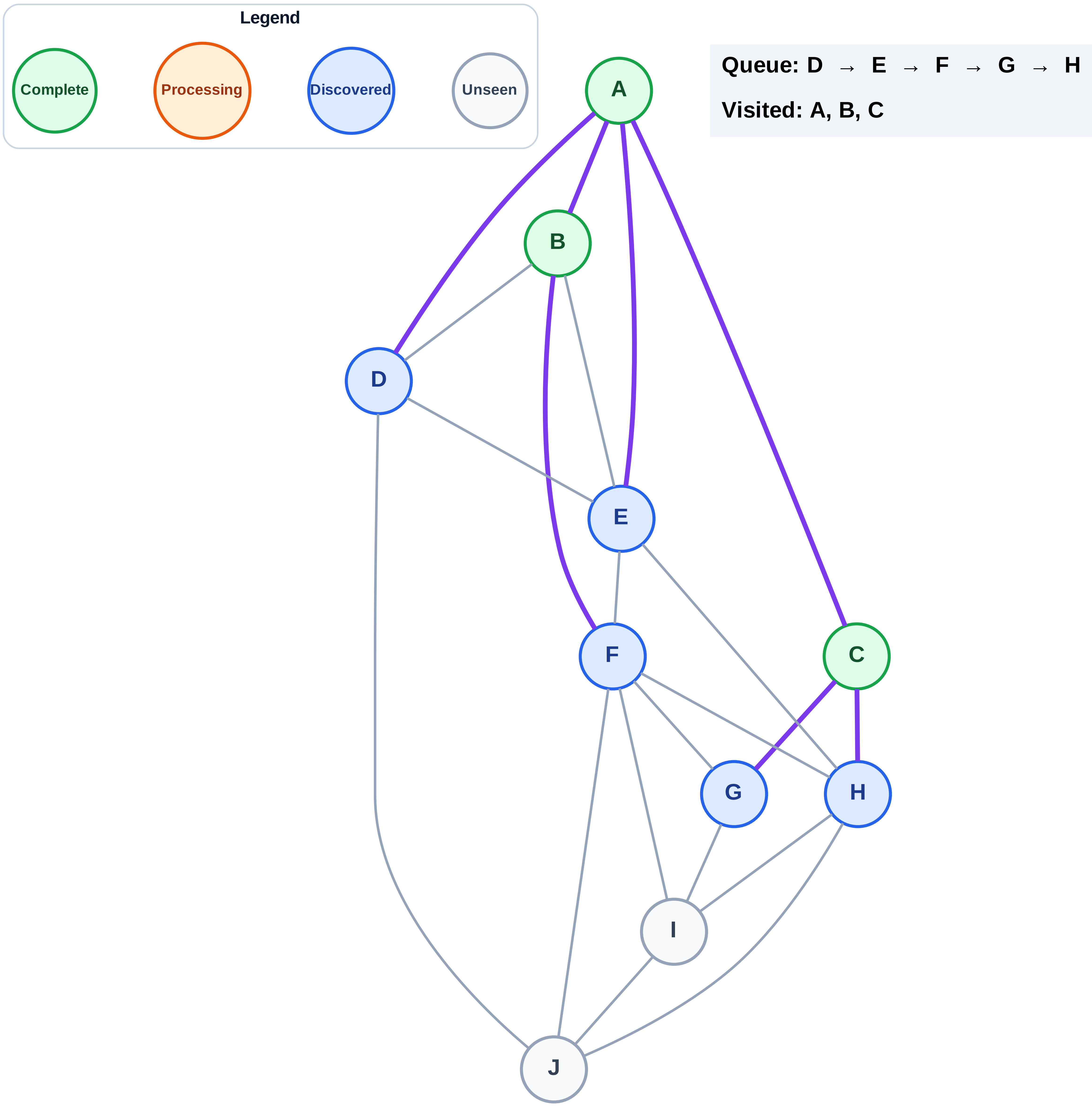
Step 5: Discover F from B




BFS Traversal
Step 6: Process node C
Legend
Complete
Processing
Discovered
Unseen
Queue: D → E → F
Visited: A, B
A
B
C
D
E
F
G
H
I
J

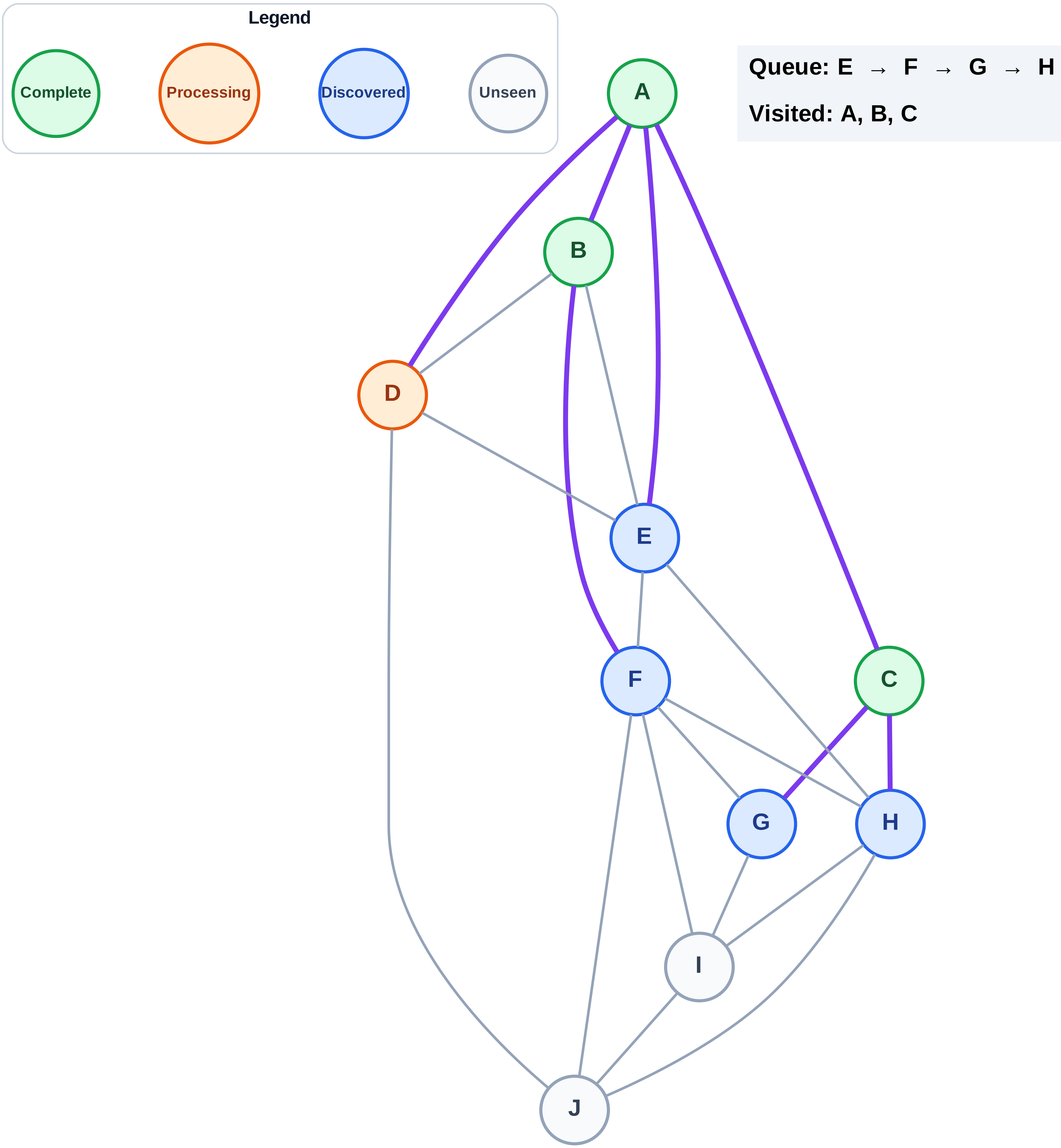
BFS Traversal

Step 7: Discover G, H from C



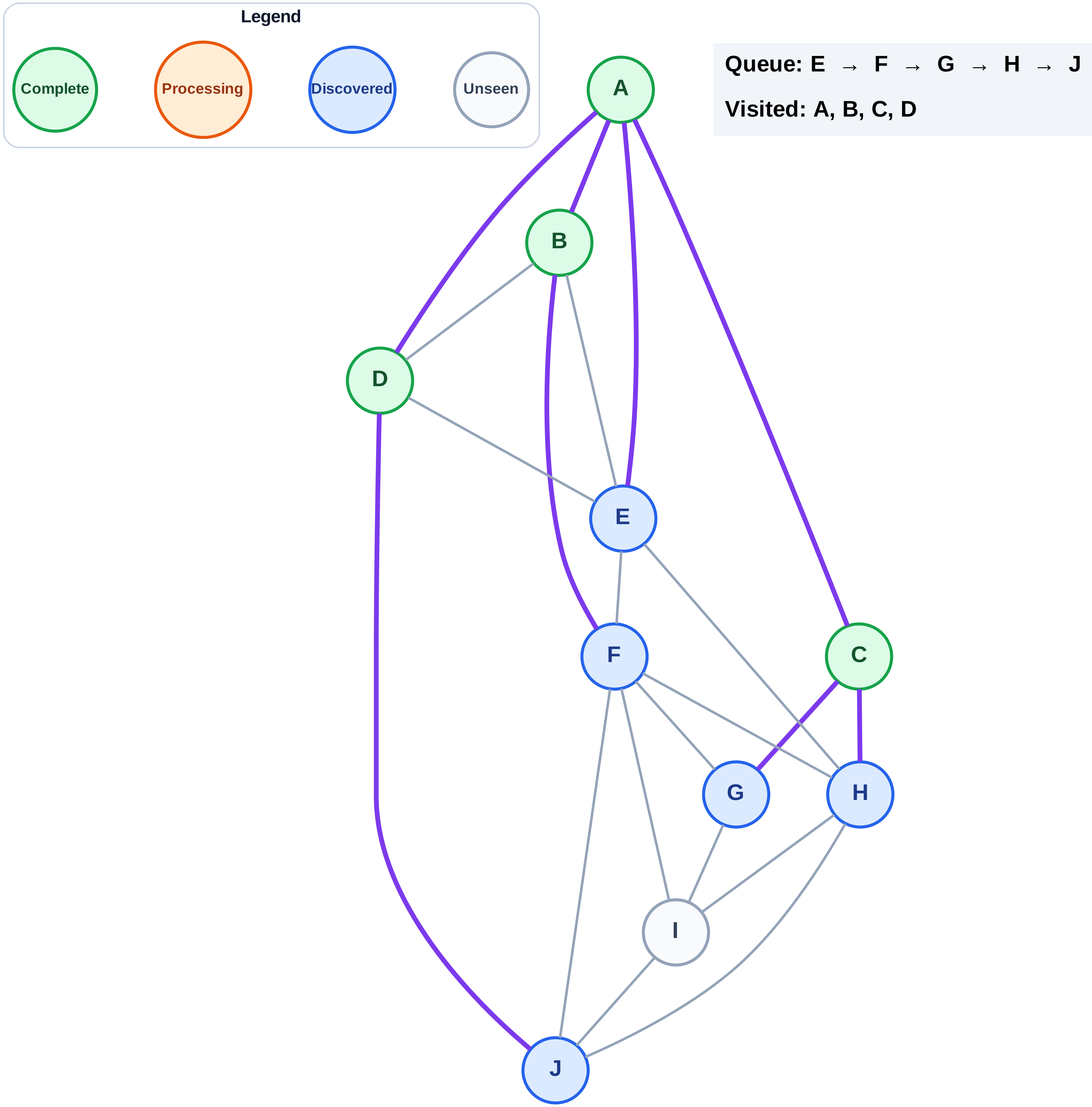
BFS Traversal

Step 8: Process node D



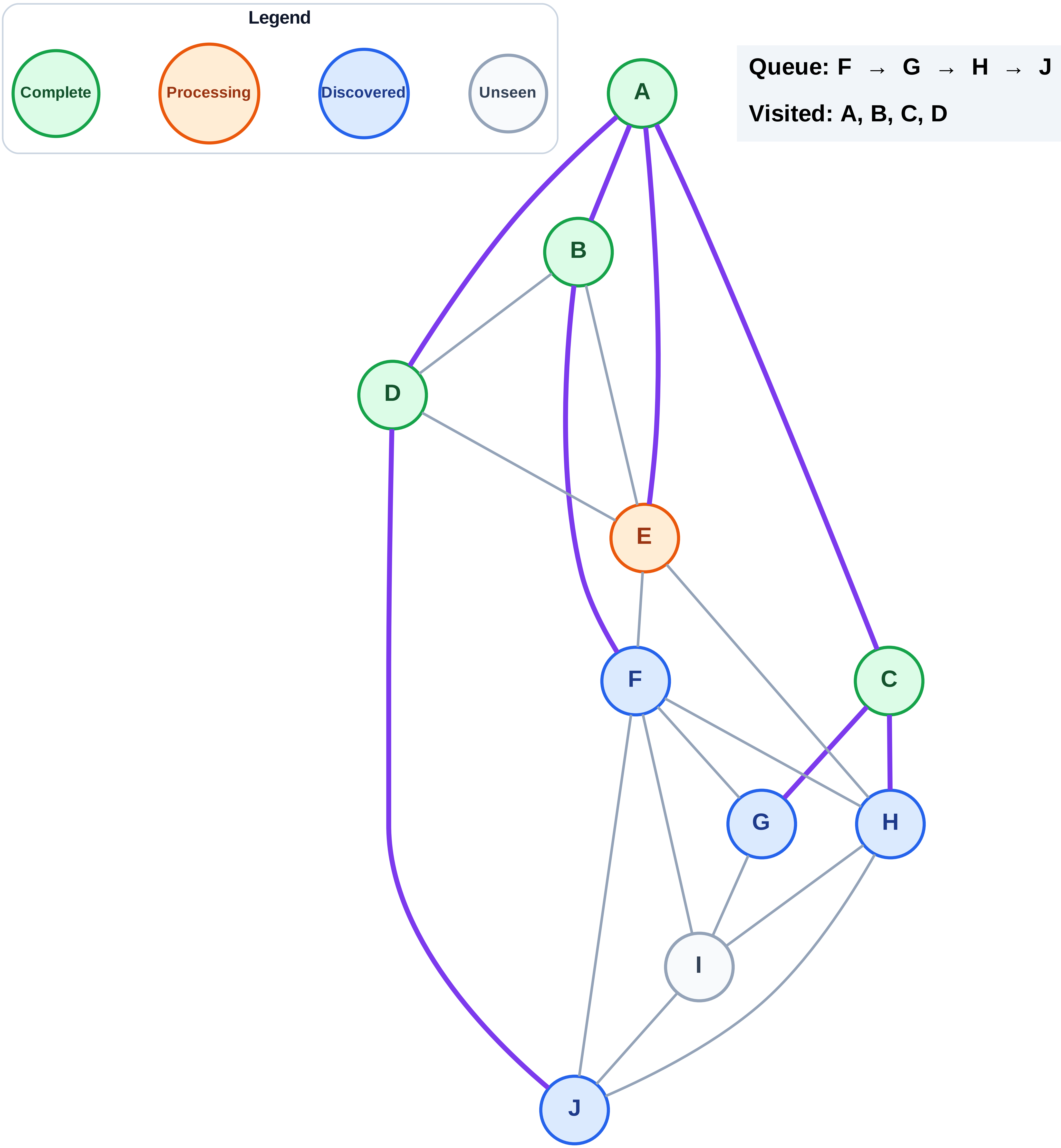
BFS Traversal

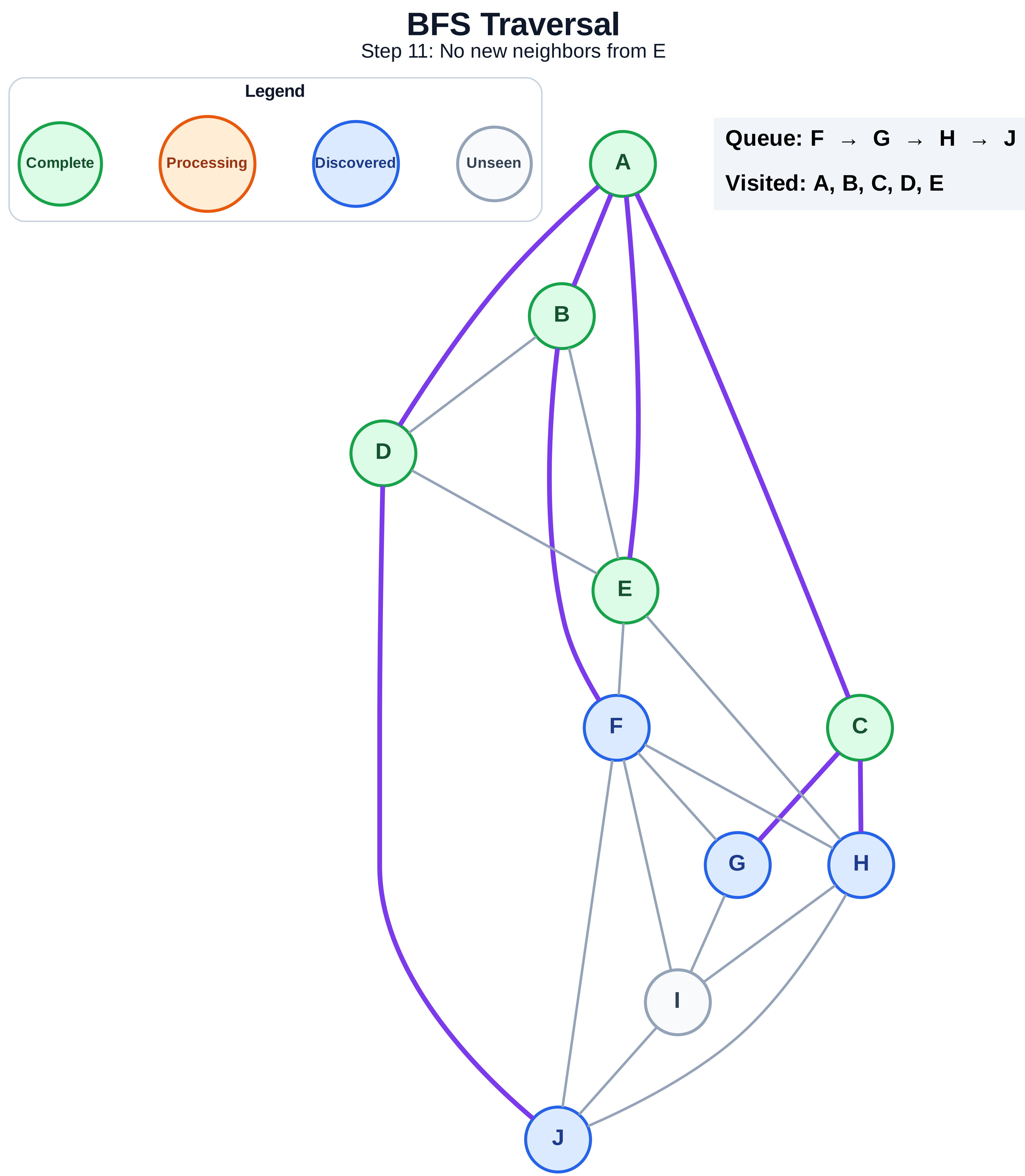
Step 9: Discover J from D



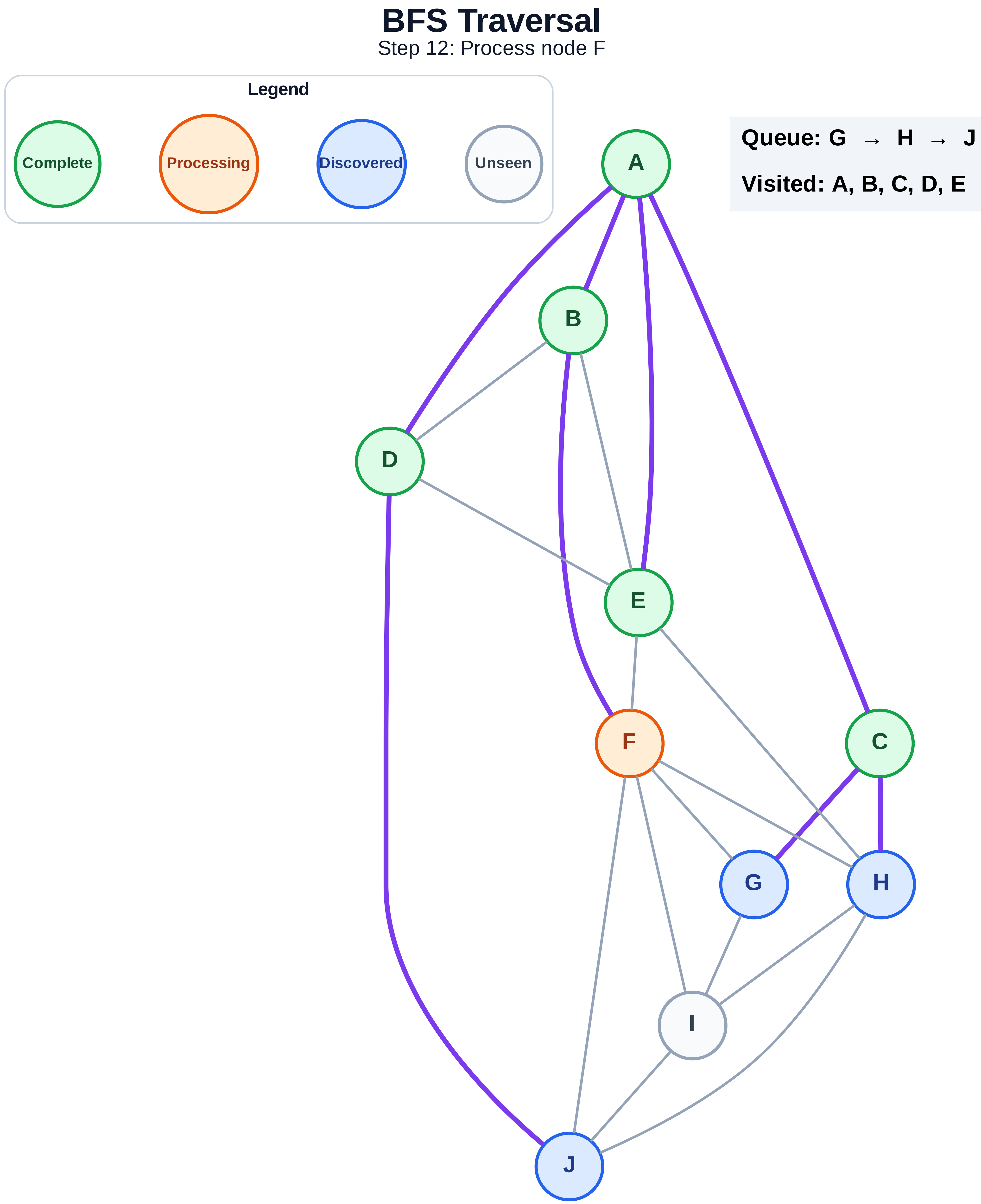
BFS Traversal

Step 10: Process node E





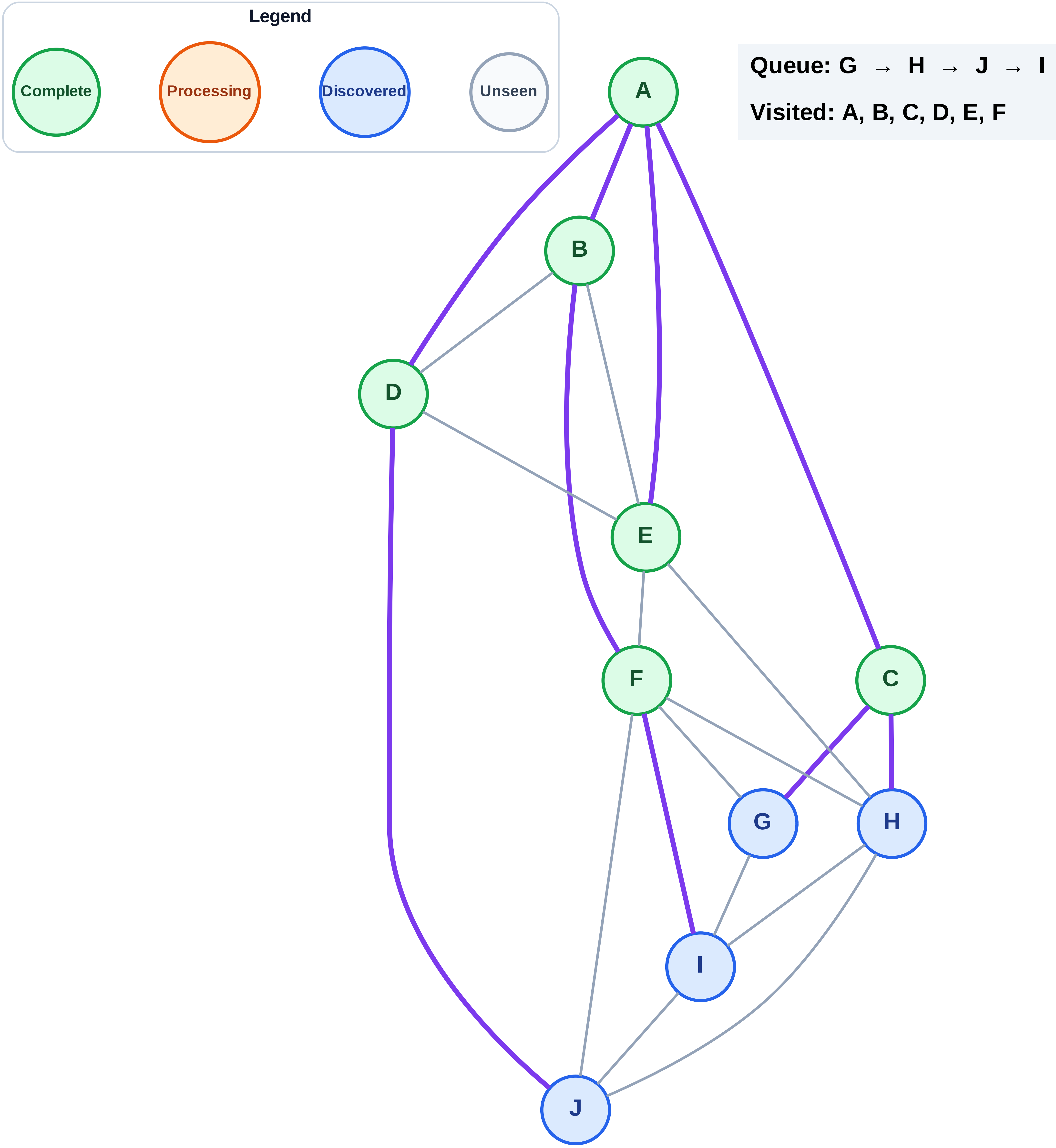
BFS Traversal
Step 11: No new neighbors from E
Legend
Complete
Processing
Discovered
Unseen
Queue: F → G → H → J
Visited: A, B, C, D, E
A
B
C
D
E
F
G
H
I
J



BFS Traversal
Step 12: Process node F
Legend
Complete
Processing
Discovered
Unseen
Queue: G → H → J
Visited: A, B, C, D, E
A
B
C
D
E
F
G
H
I
J

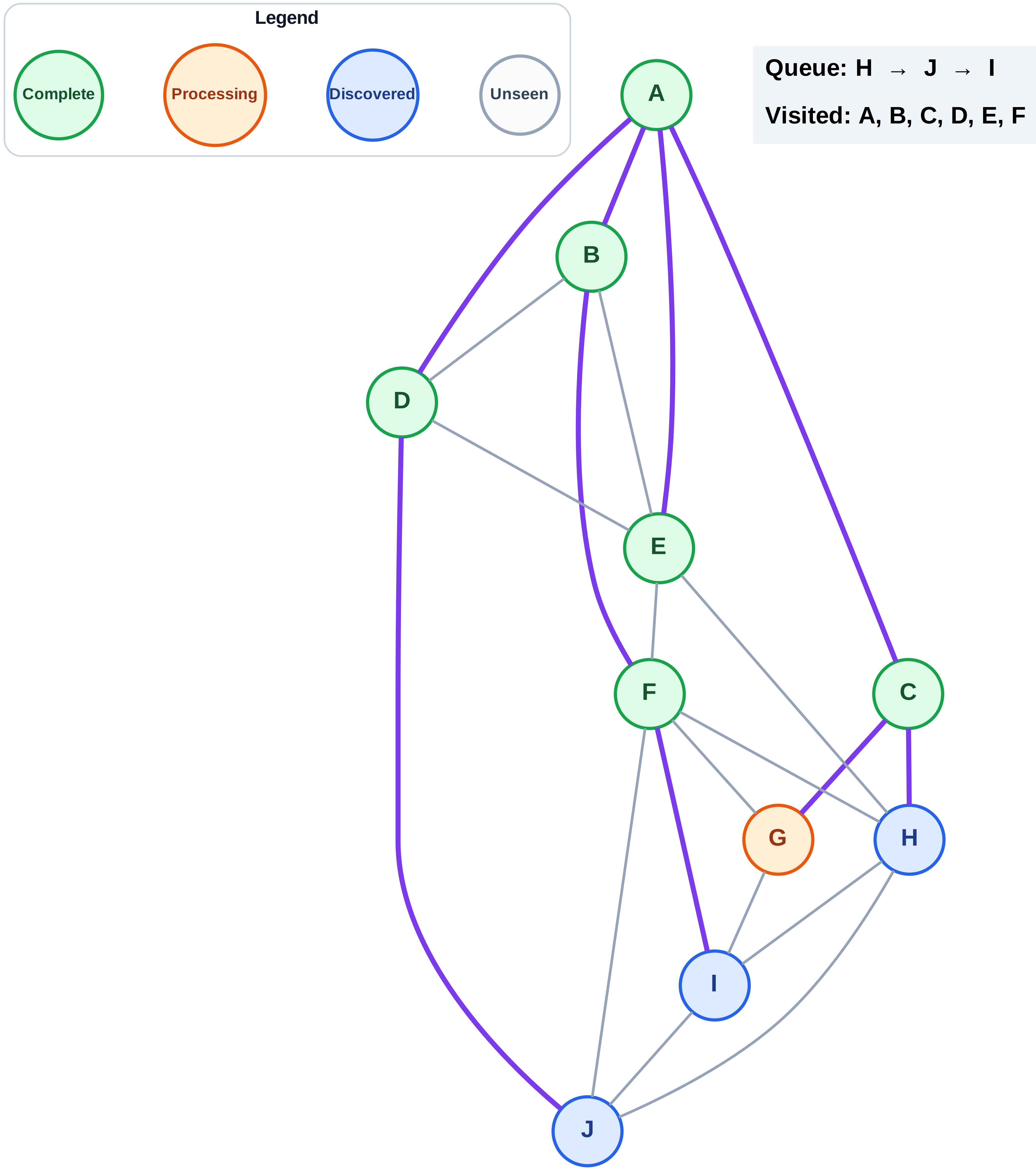
BFS Traversal

Step 13: Discover I from F



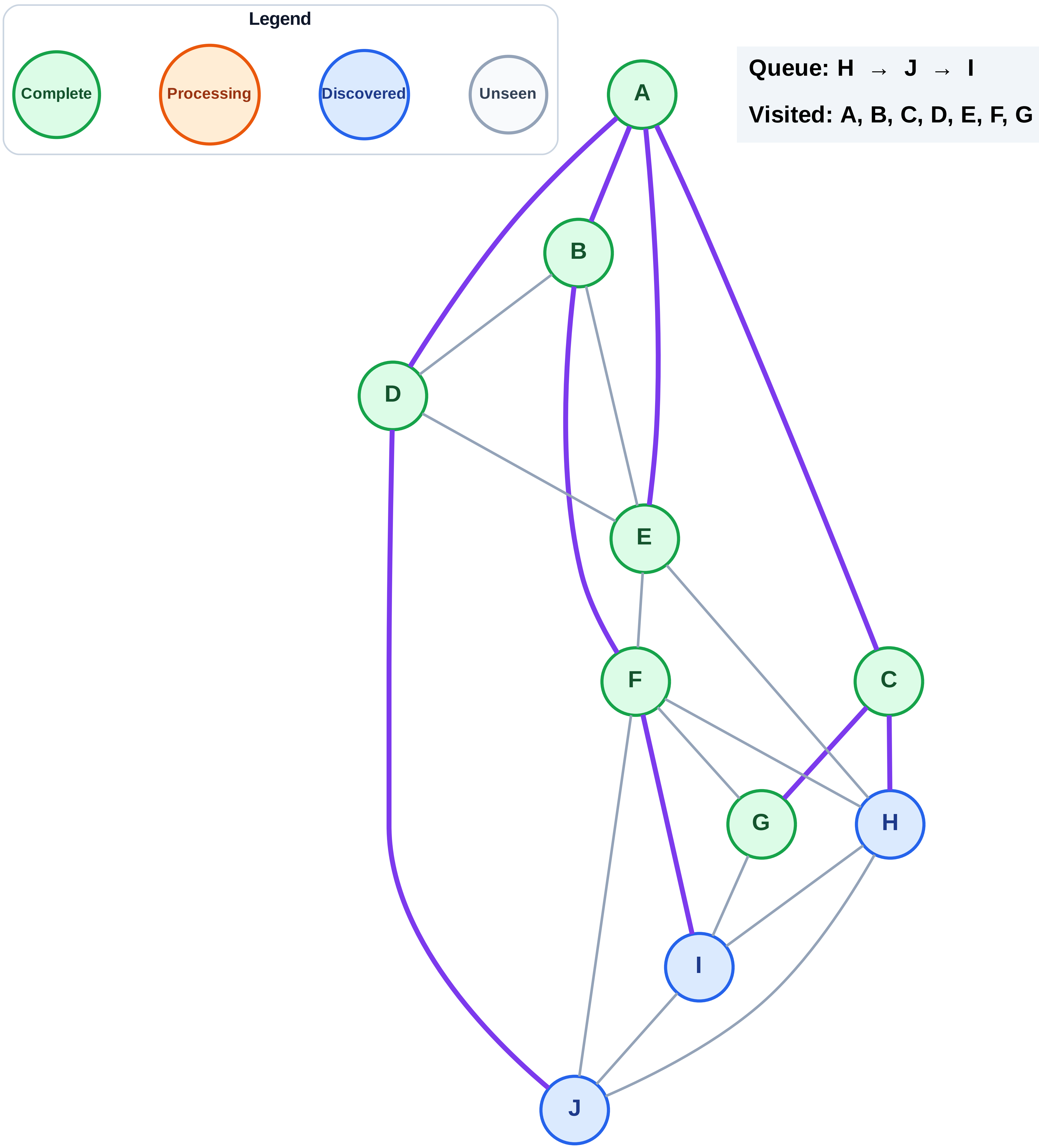
BFS Traversal

Step 14: Process node G



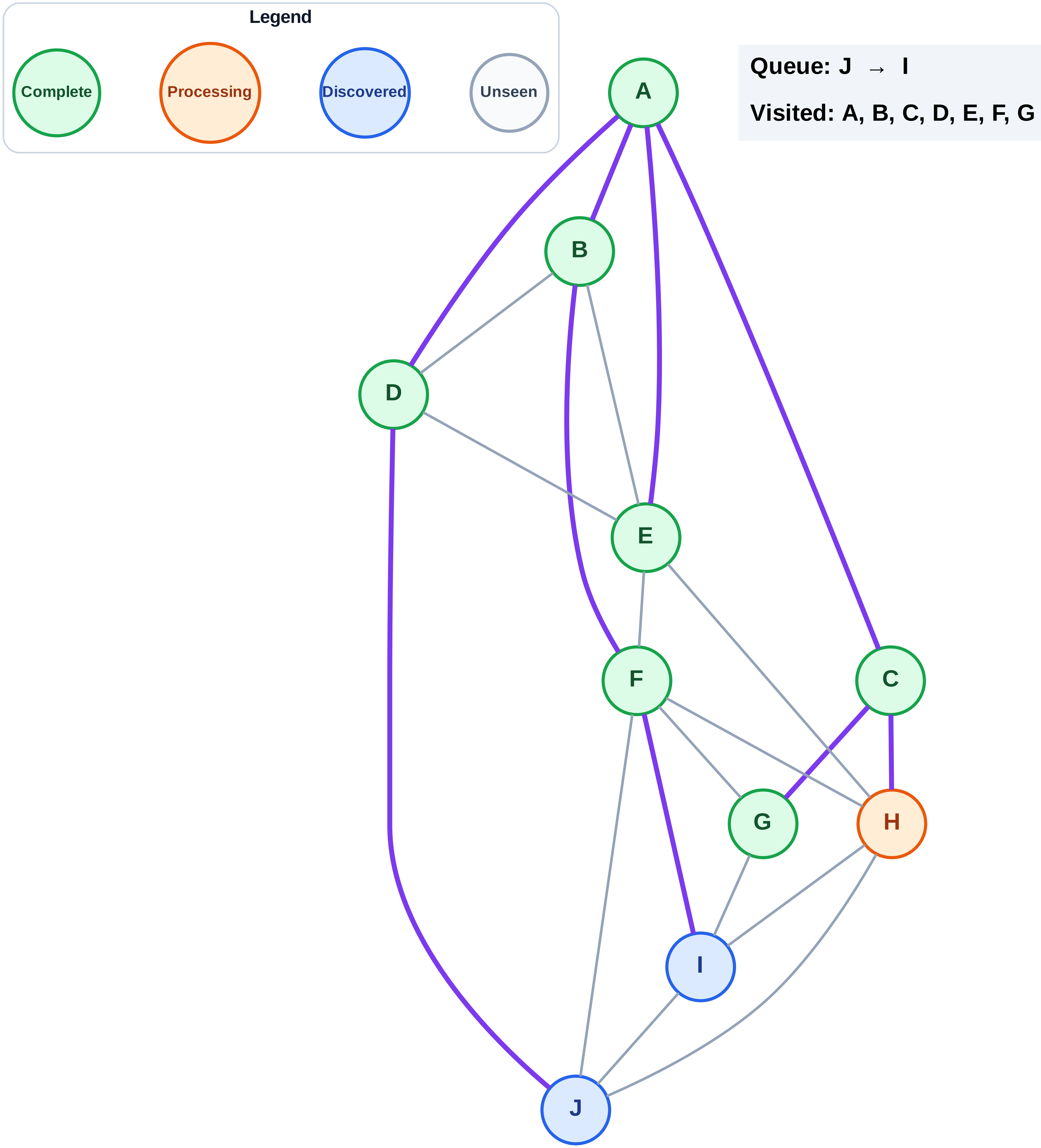
BFS Traversal

Step 15: No new neighbors from G



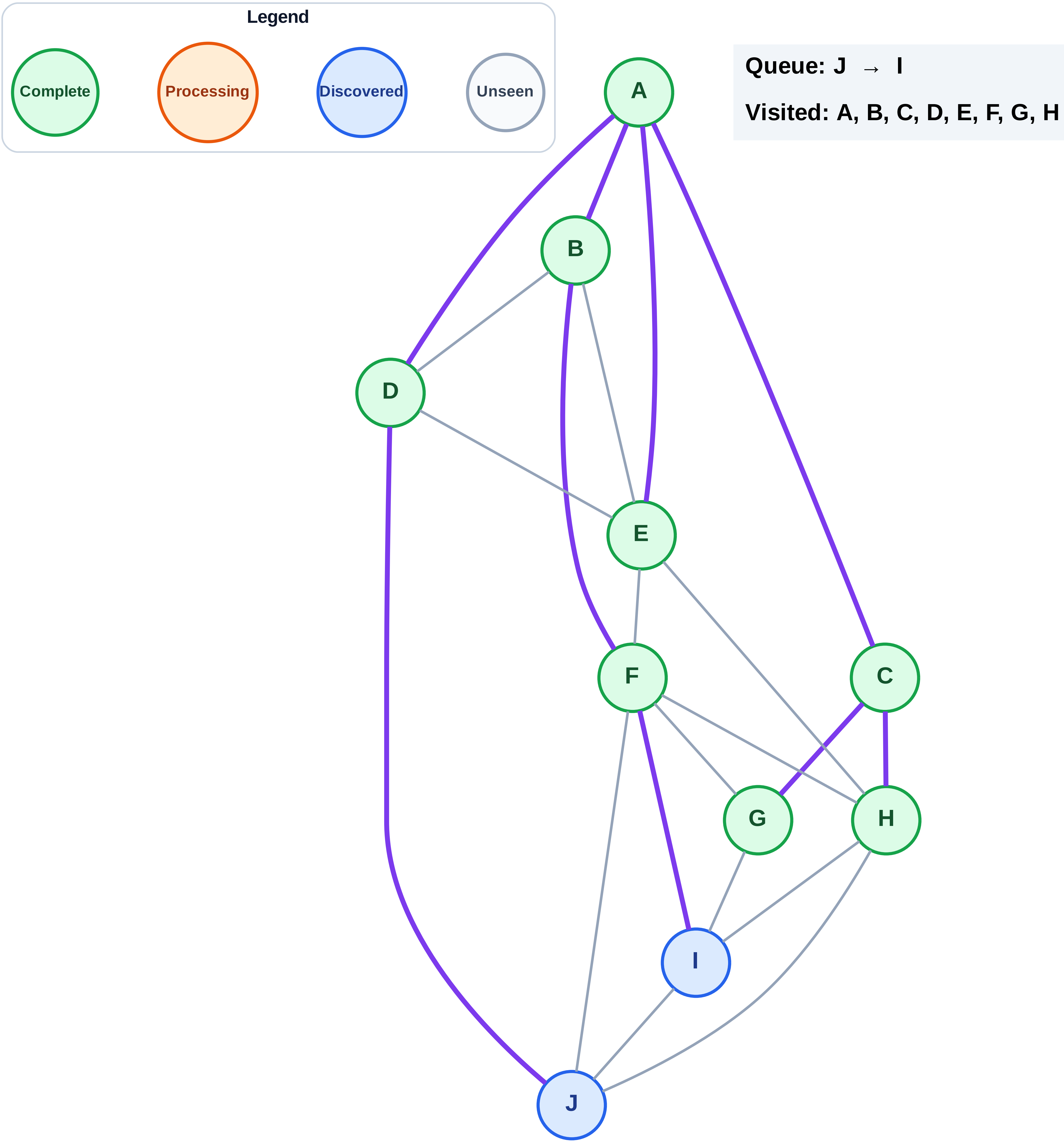
BFS Traversal

Step 16: Process node H



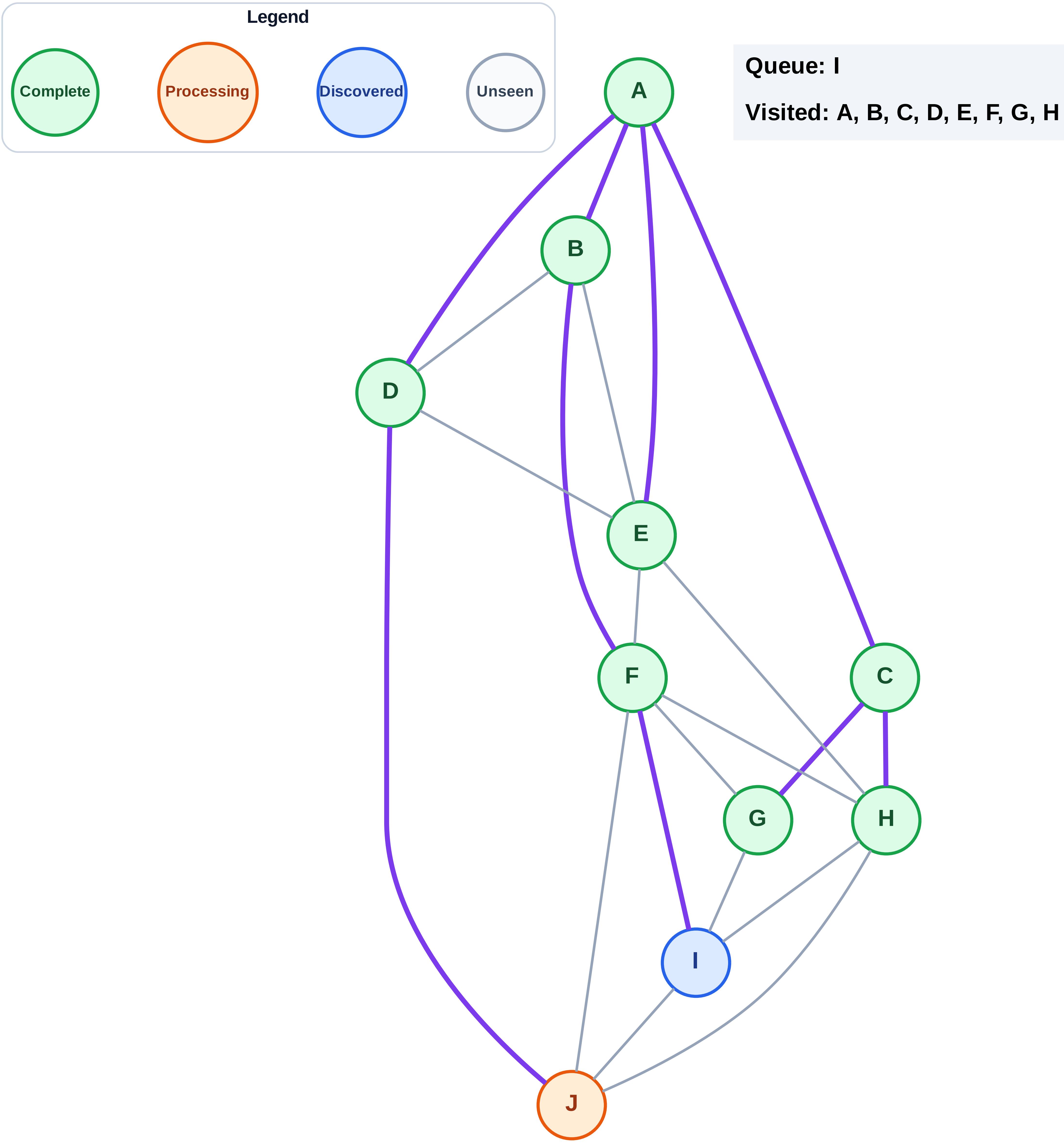
BFS Traversal

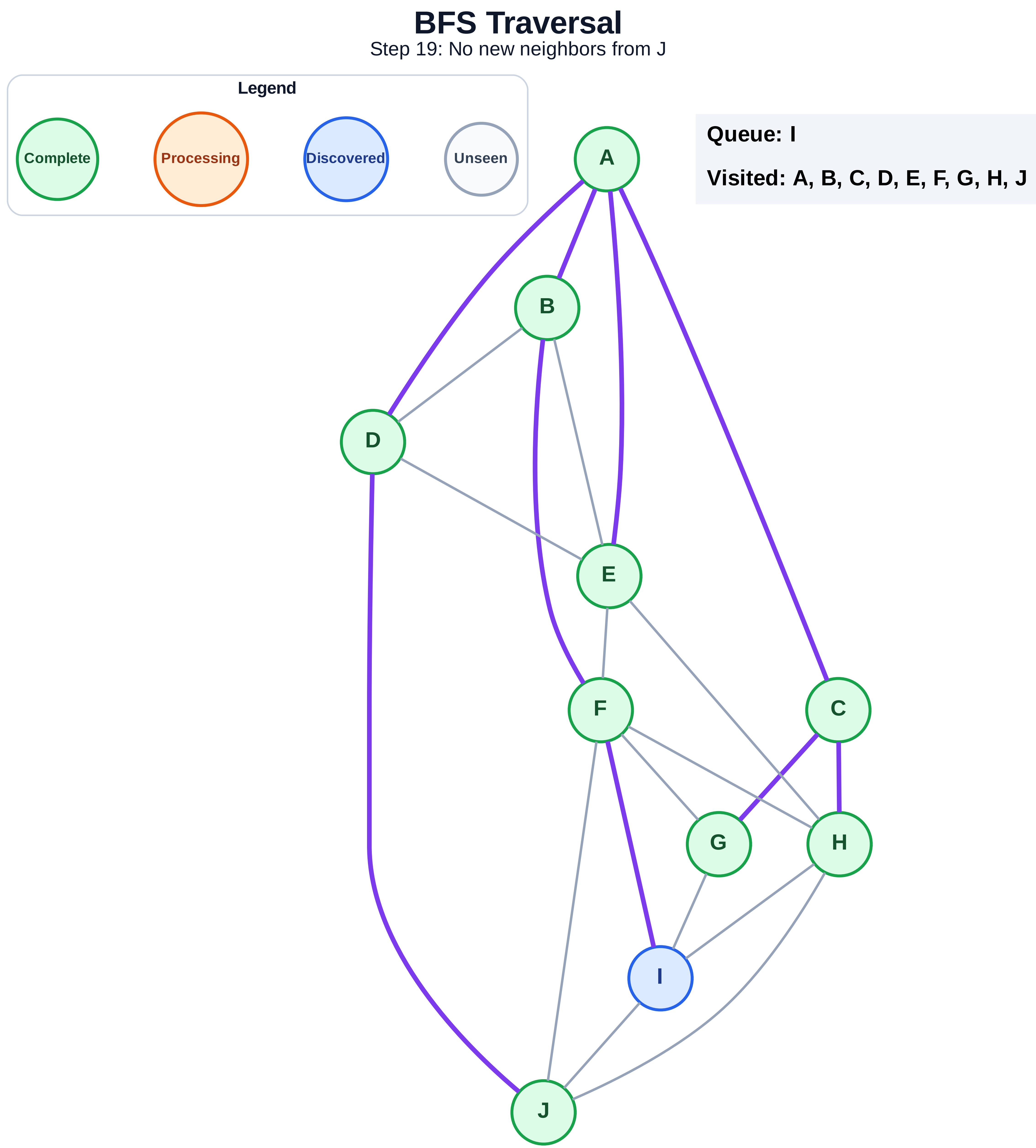
Step 17: No new neighbors from H



BFS Traversal

Step 18: Process node J

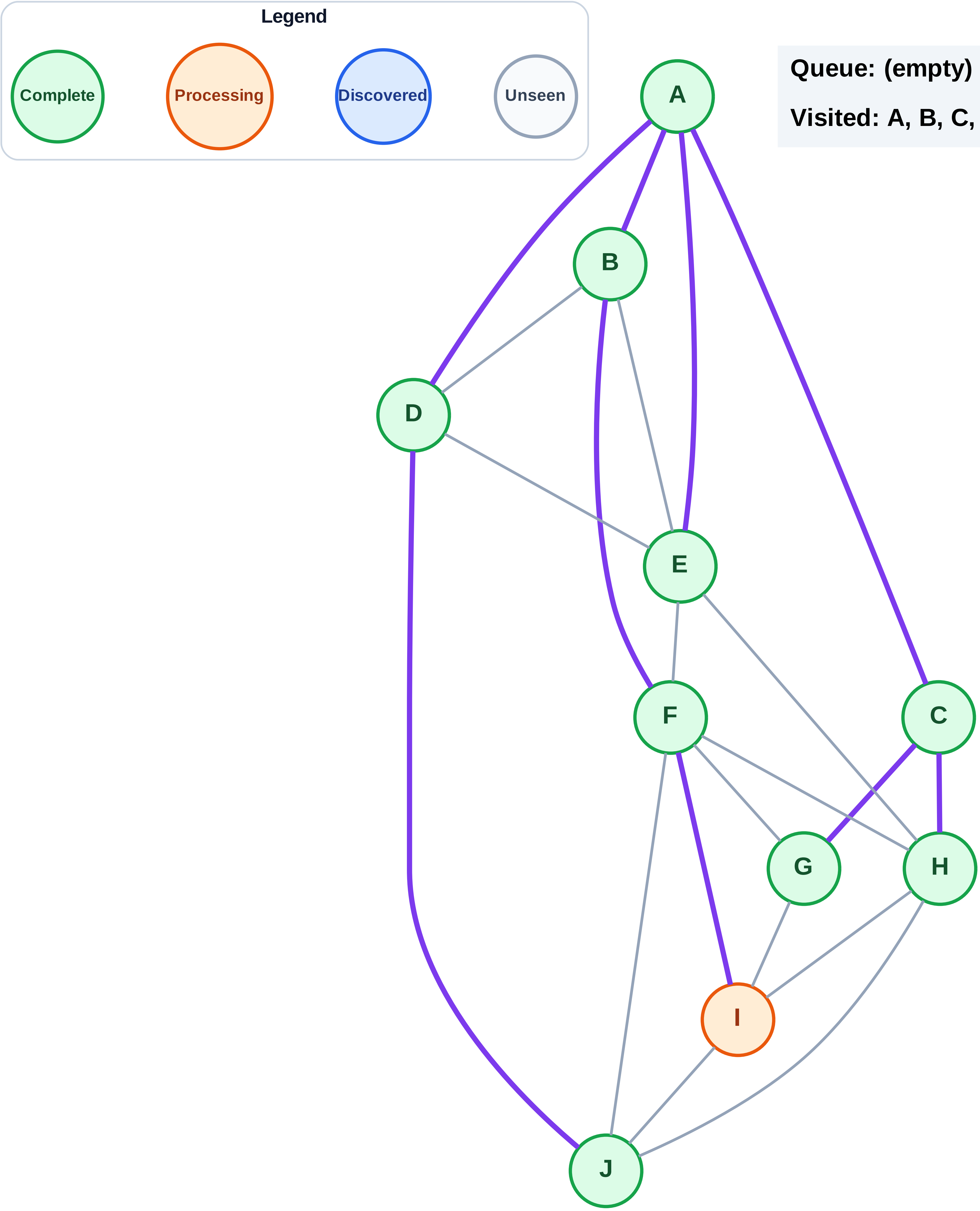




BFS Traversal
Step 19: No new neighbors from J
Legend
Complete
Processing
Discovered
Unseen
Queue: I
Visited: A, B, C, D, E, F, G, H, J
A
B
C
D
E
F
G
H
I
J

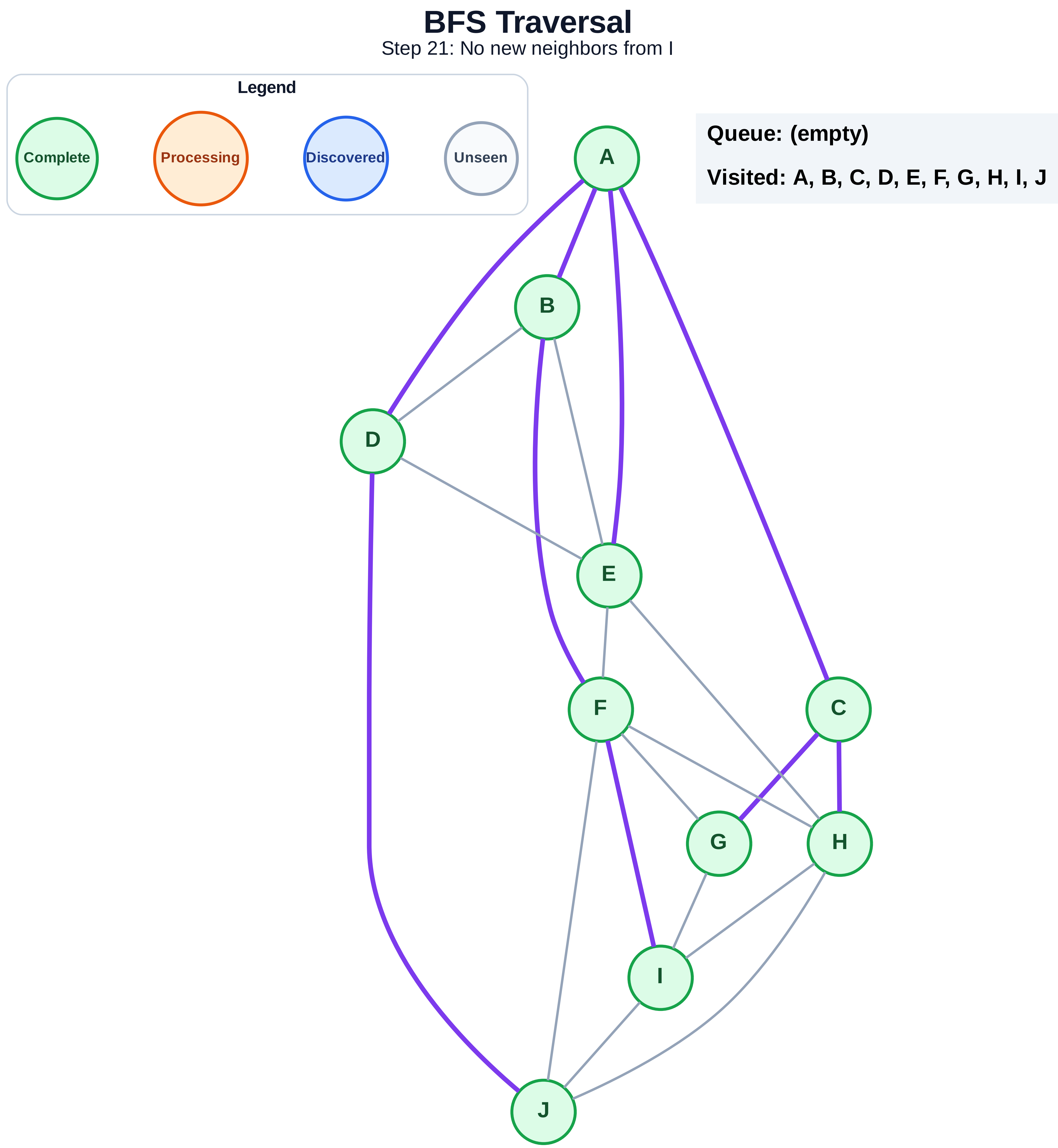
BFS Traversal

Step 20: Process node I



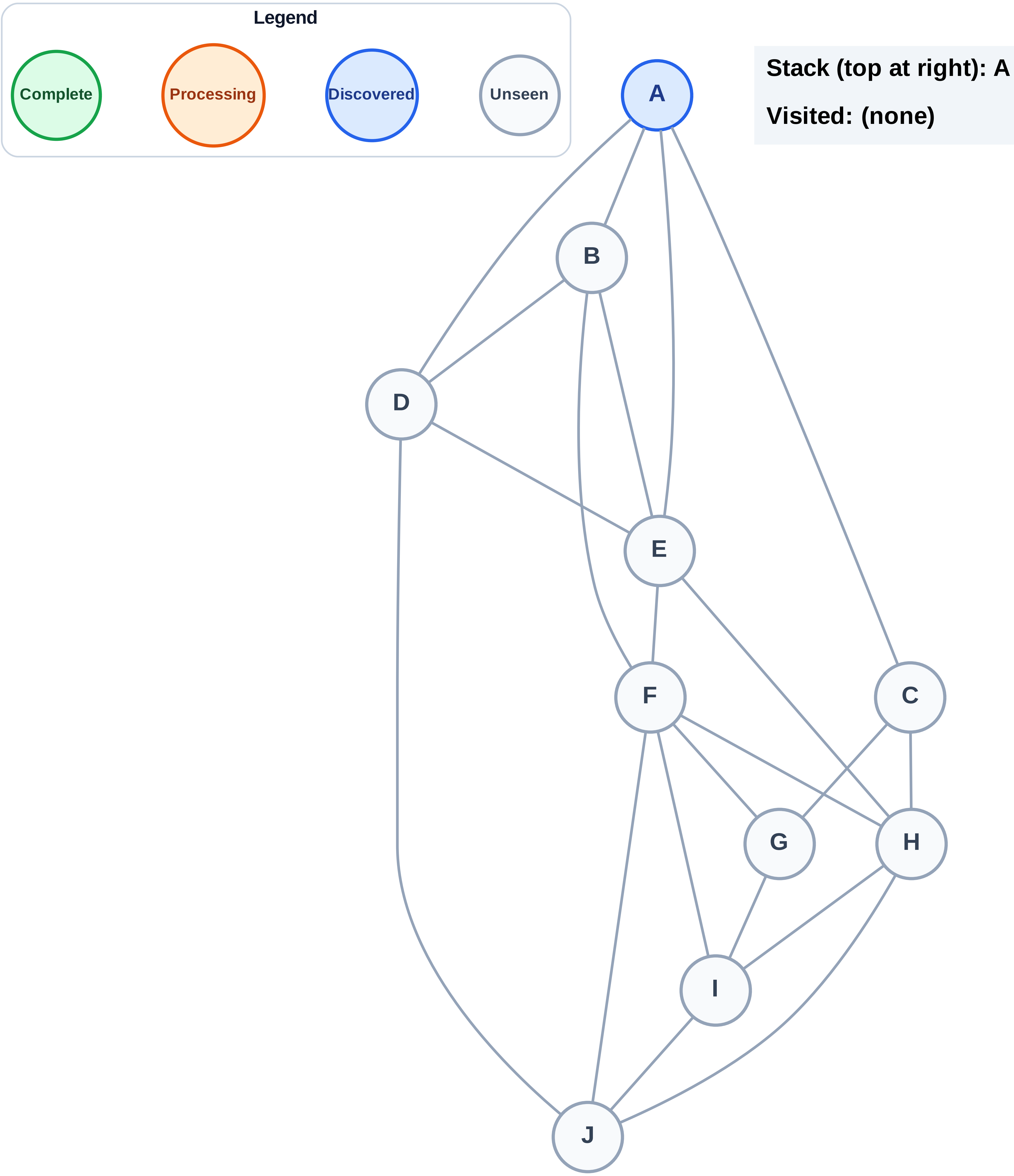
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J


BFS Traversal
Step 21: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J
A
B
C
D
E
F
G
H
I
J

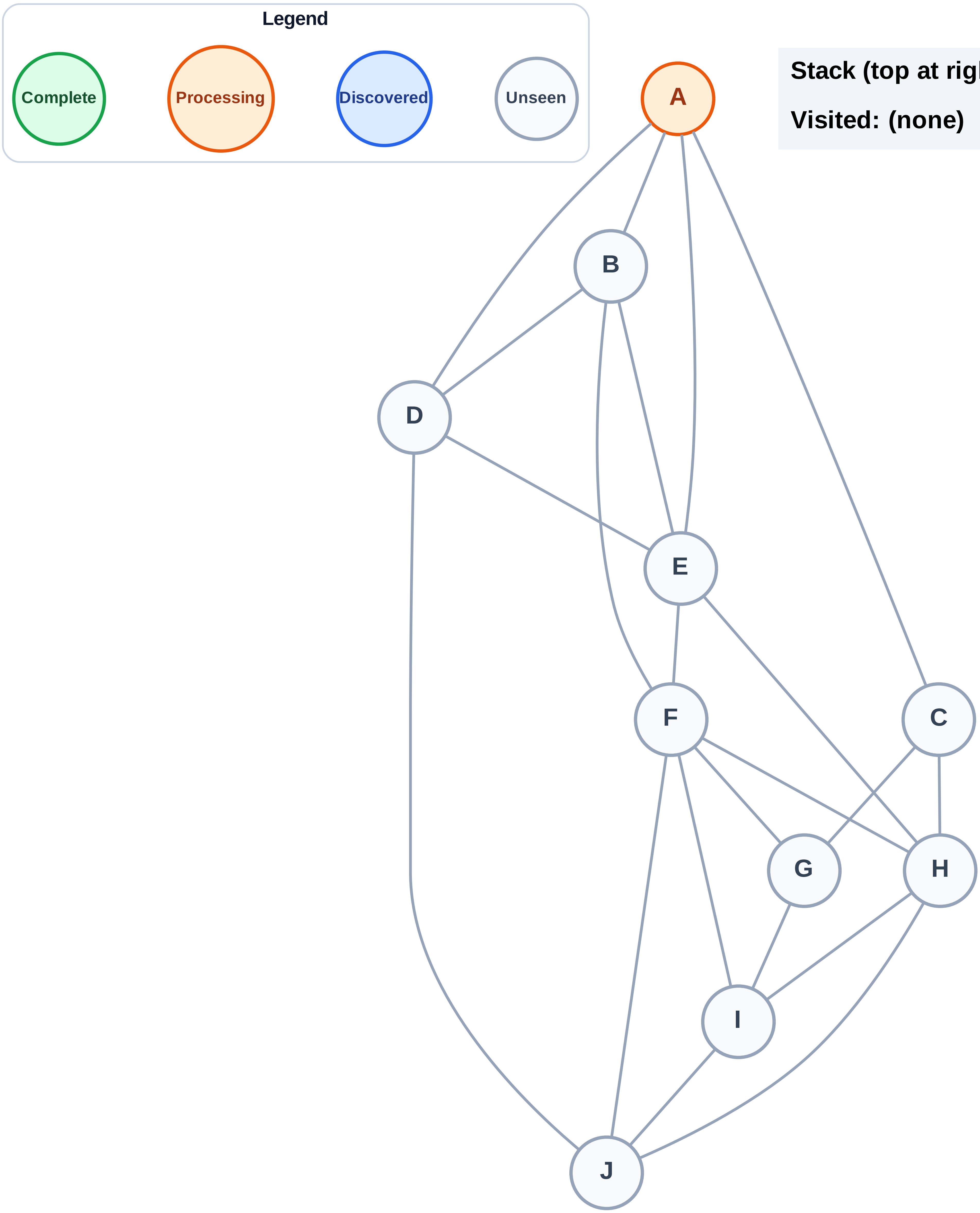
DFS Traversal

Step 1: Start at A



DFS Traversal

Step 2: Process node A

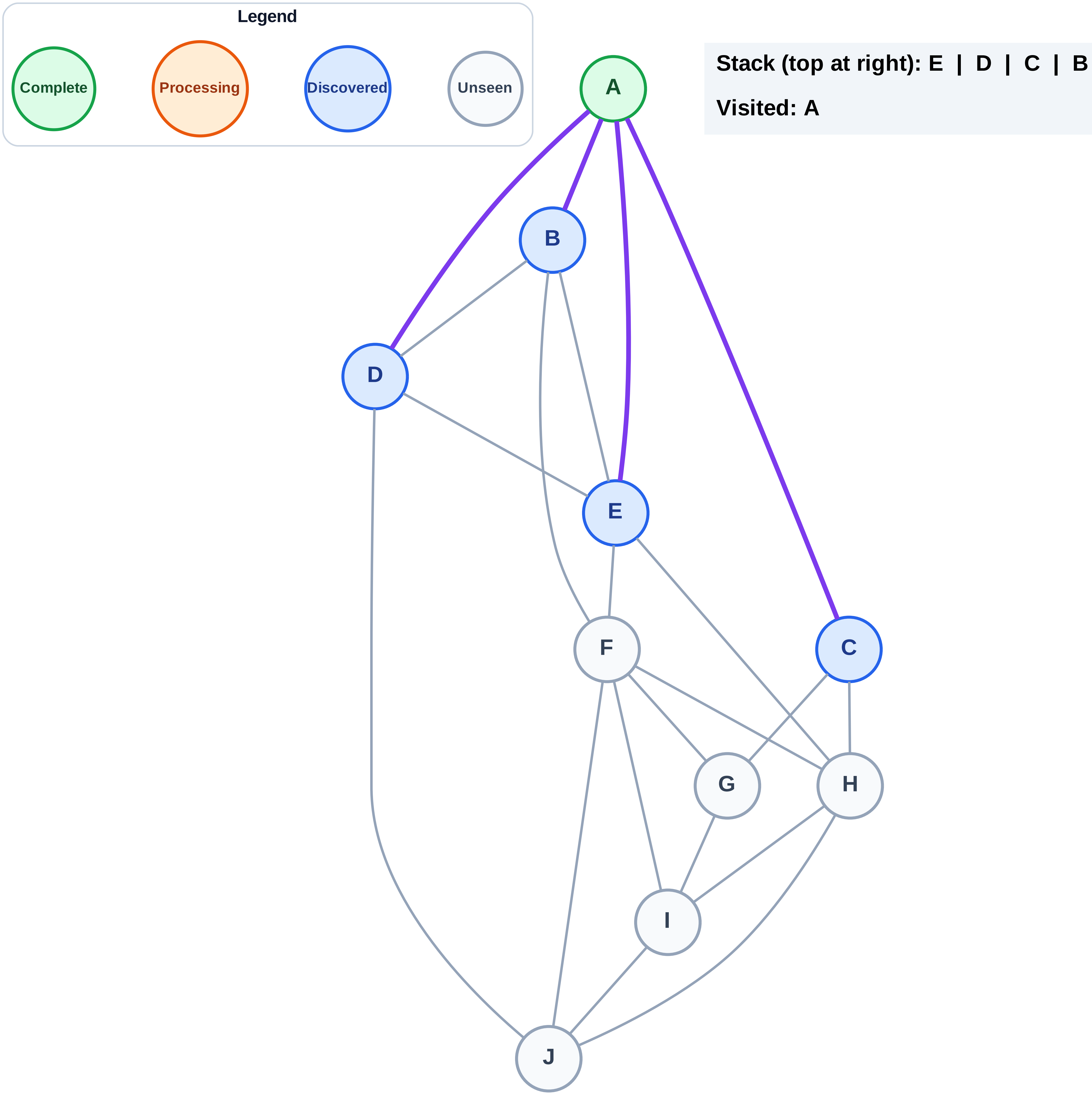


Stack (top at right): (empty)

Visited: (none)

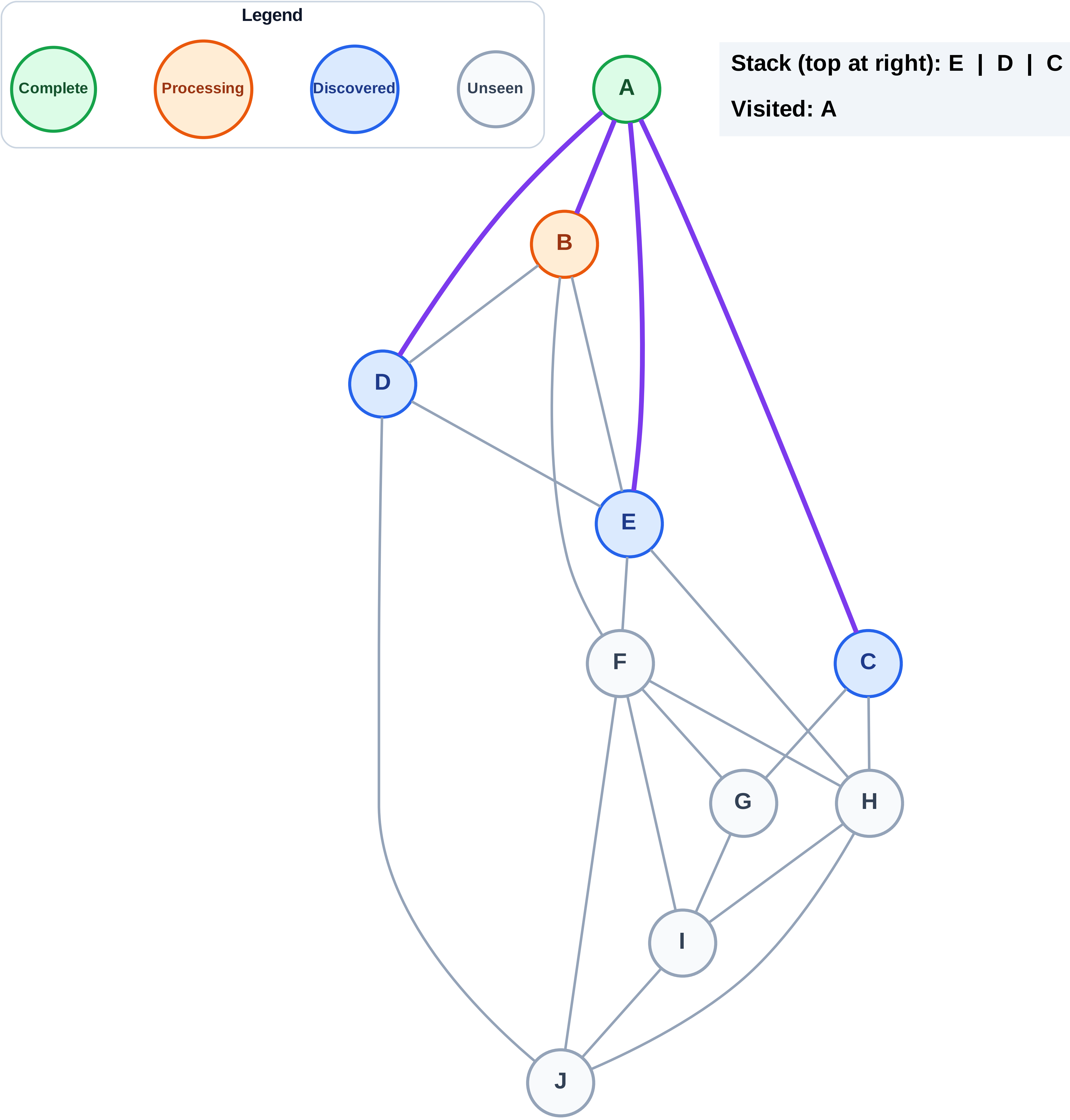
DFS Traversal

Step 3: Discover E, D, C, B from A



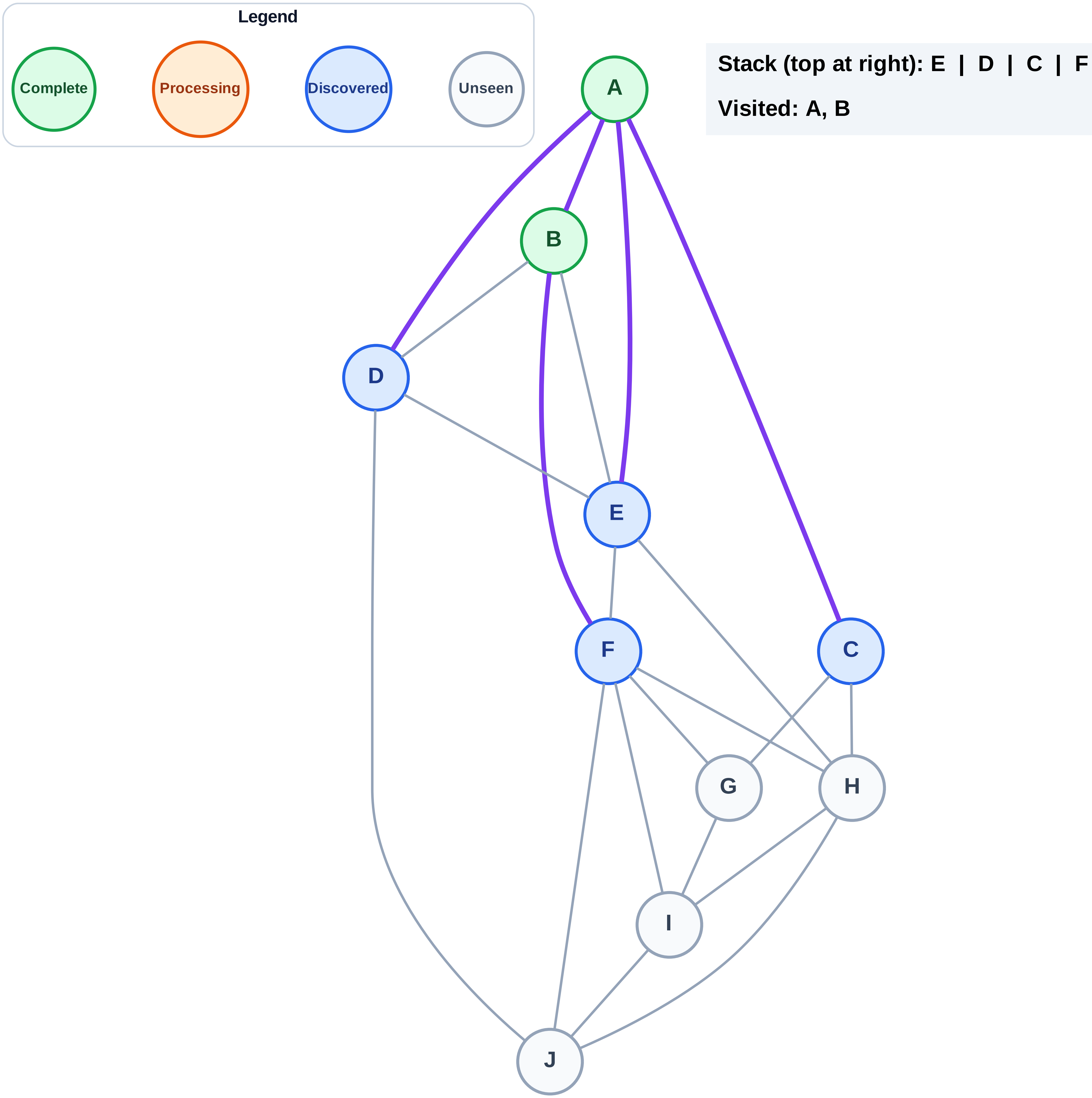
DFS Traversal

Step 4: Process node B



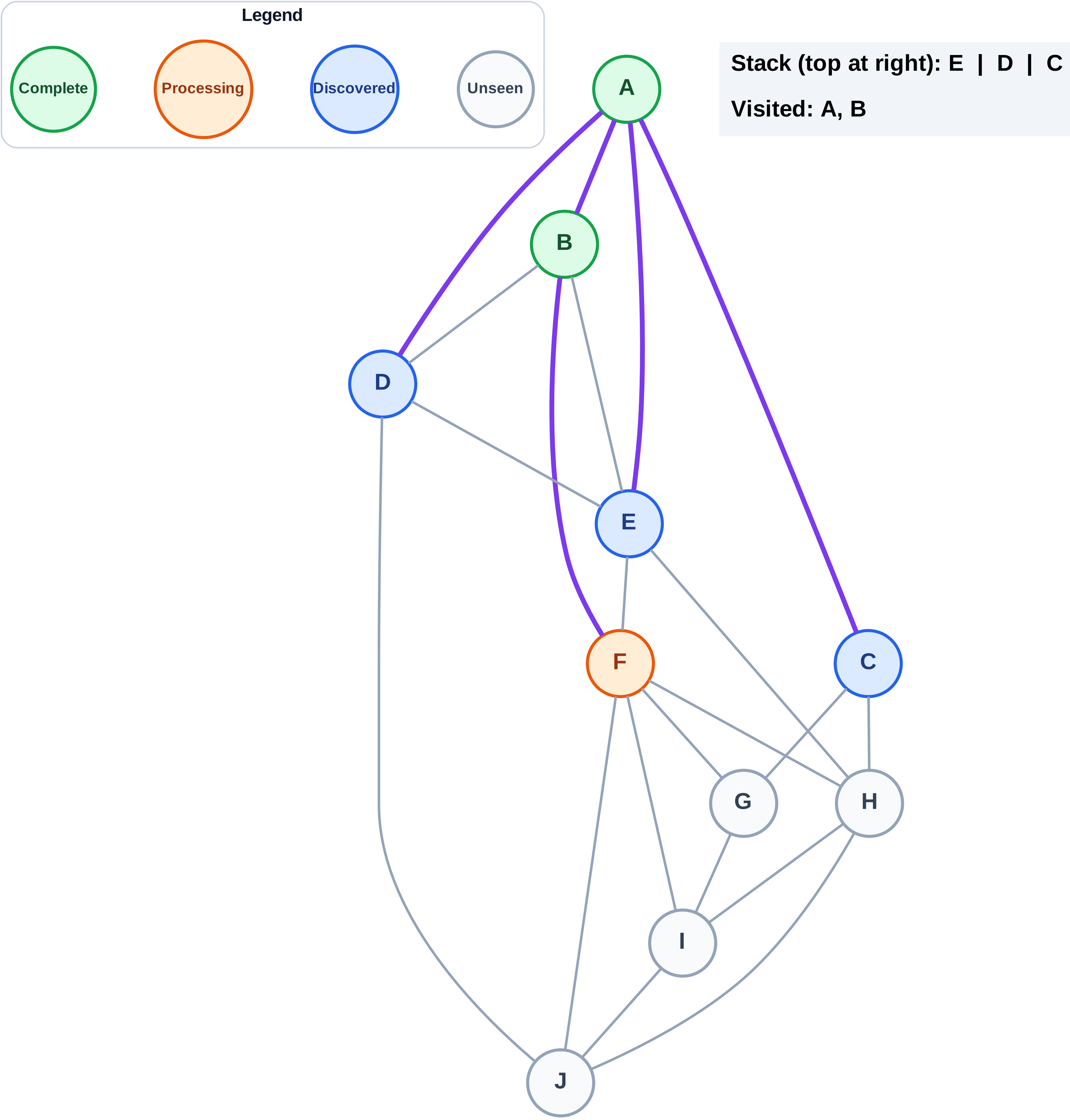
DFS Traversal

Step 5: Discover F from B



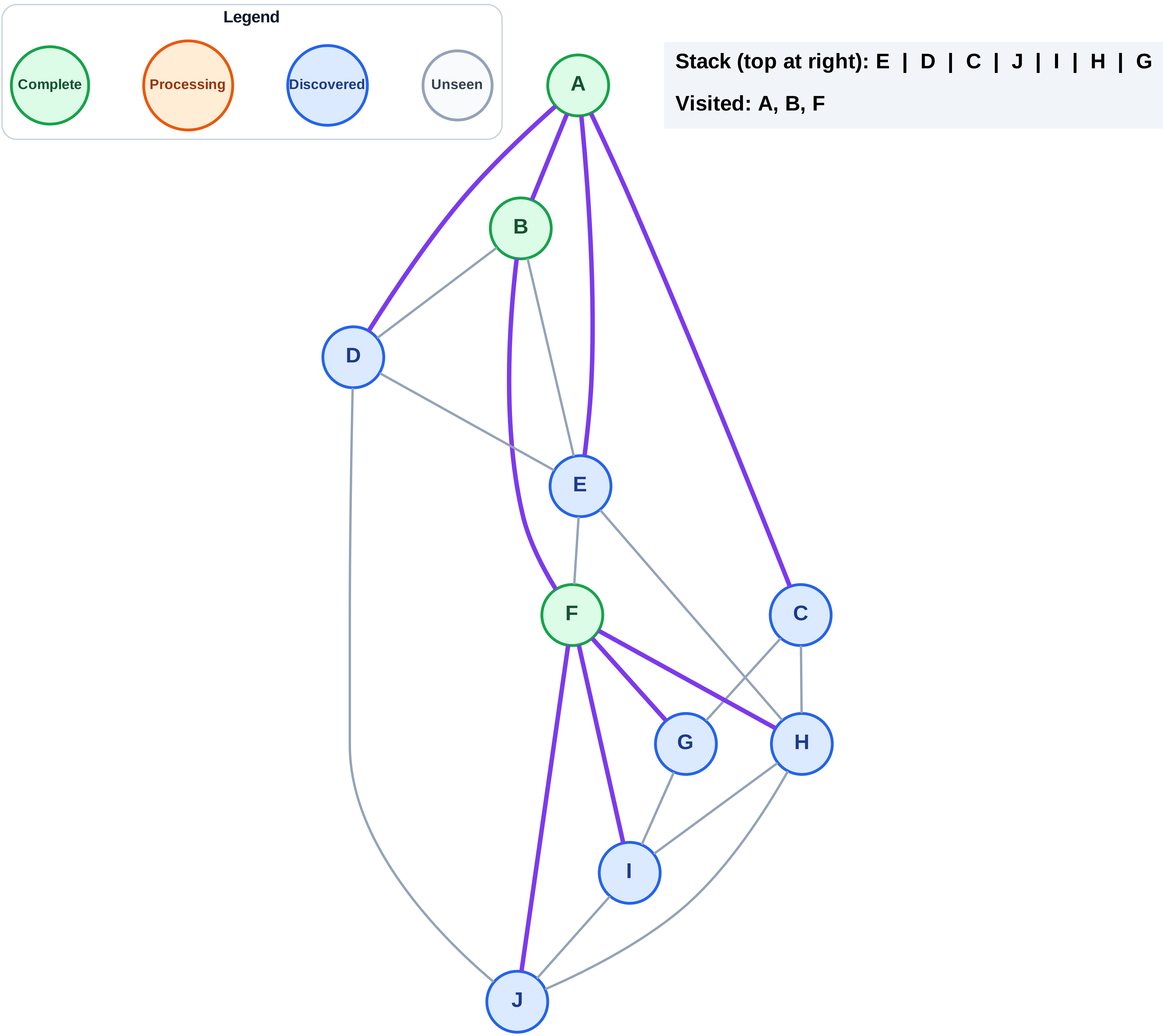
DFS Traversal

Step 6: Process node F



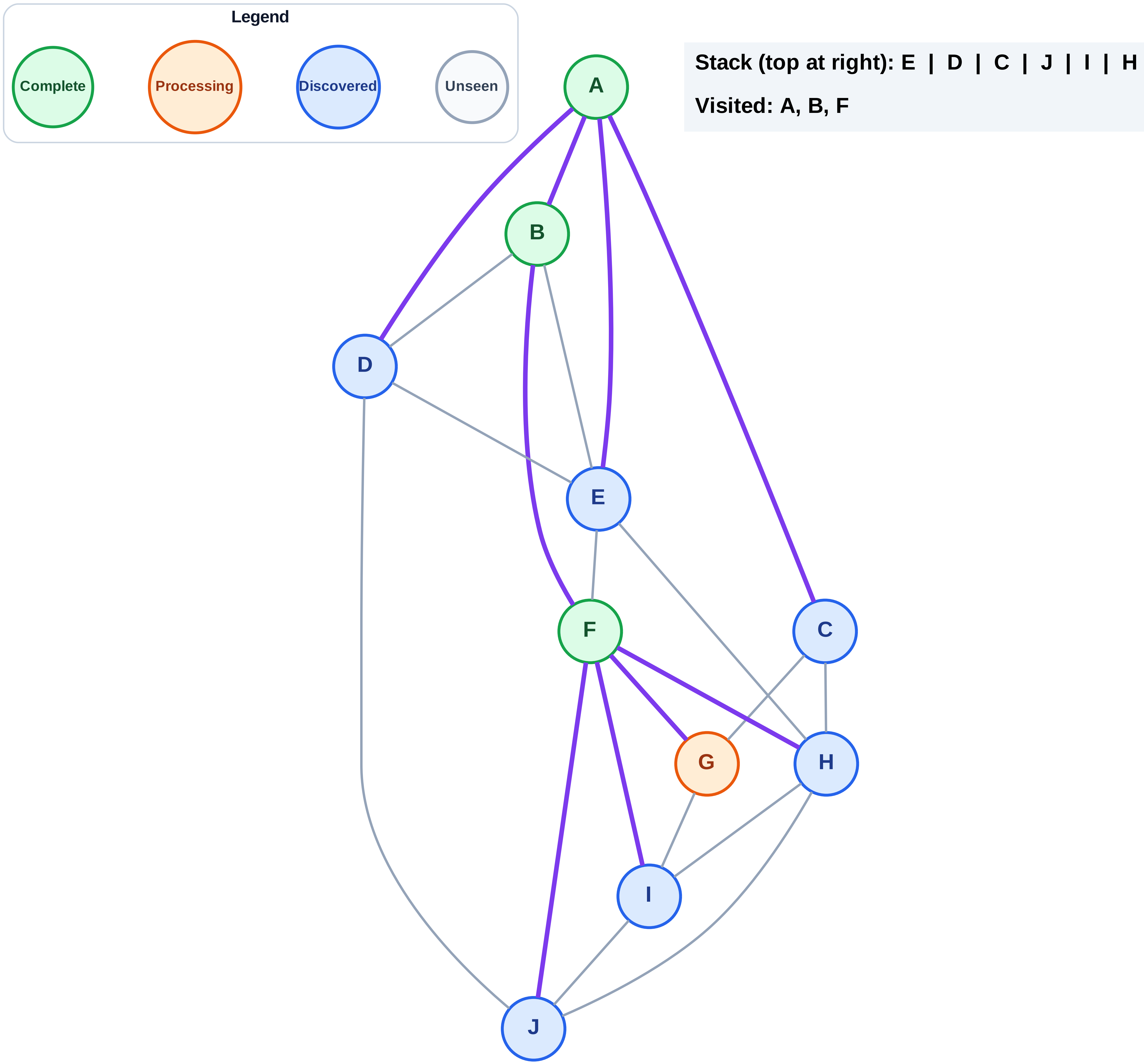
DFS Traversal

Step 7: Discover J, I, H, G from F



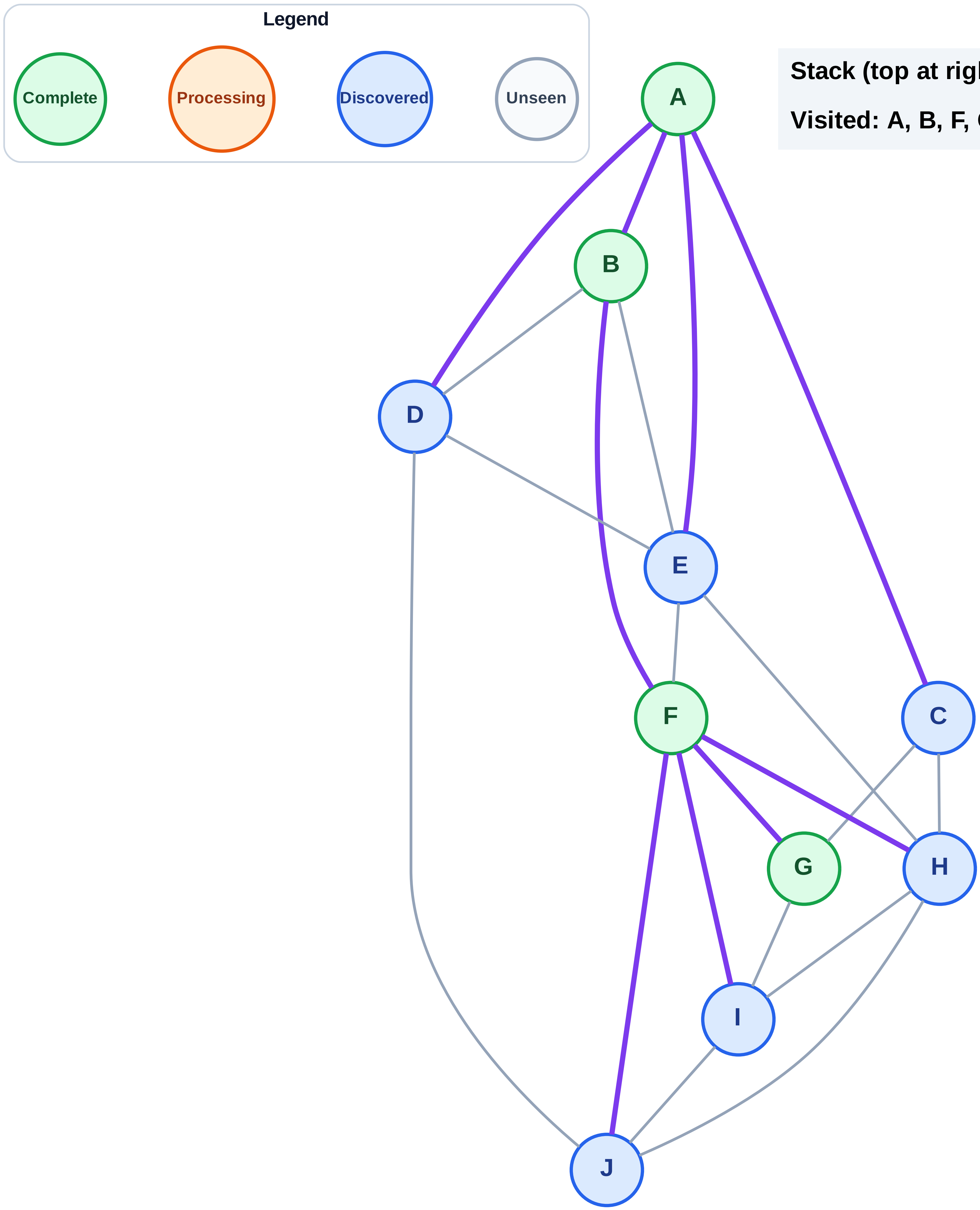
DFS Traversal

Step 8: Process node G



DFS Traversal

Step 9: No new neighbors from G

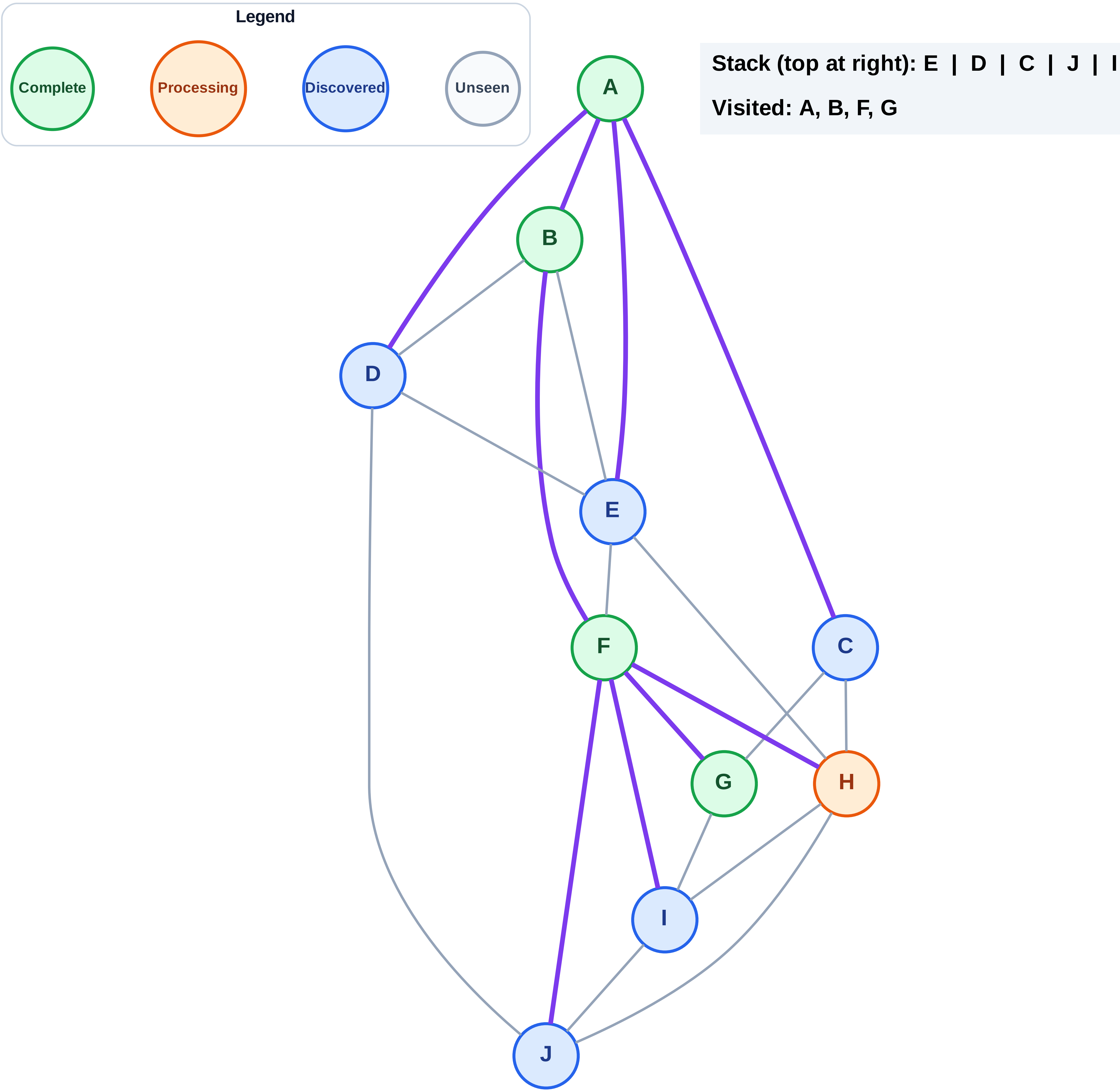


Stack (top at right): E | D | C | J | I | H

Visited: A, B, F, G

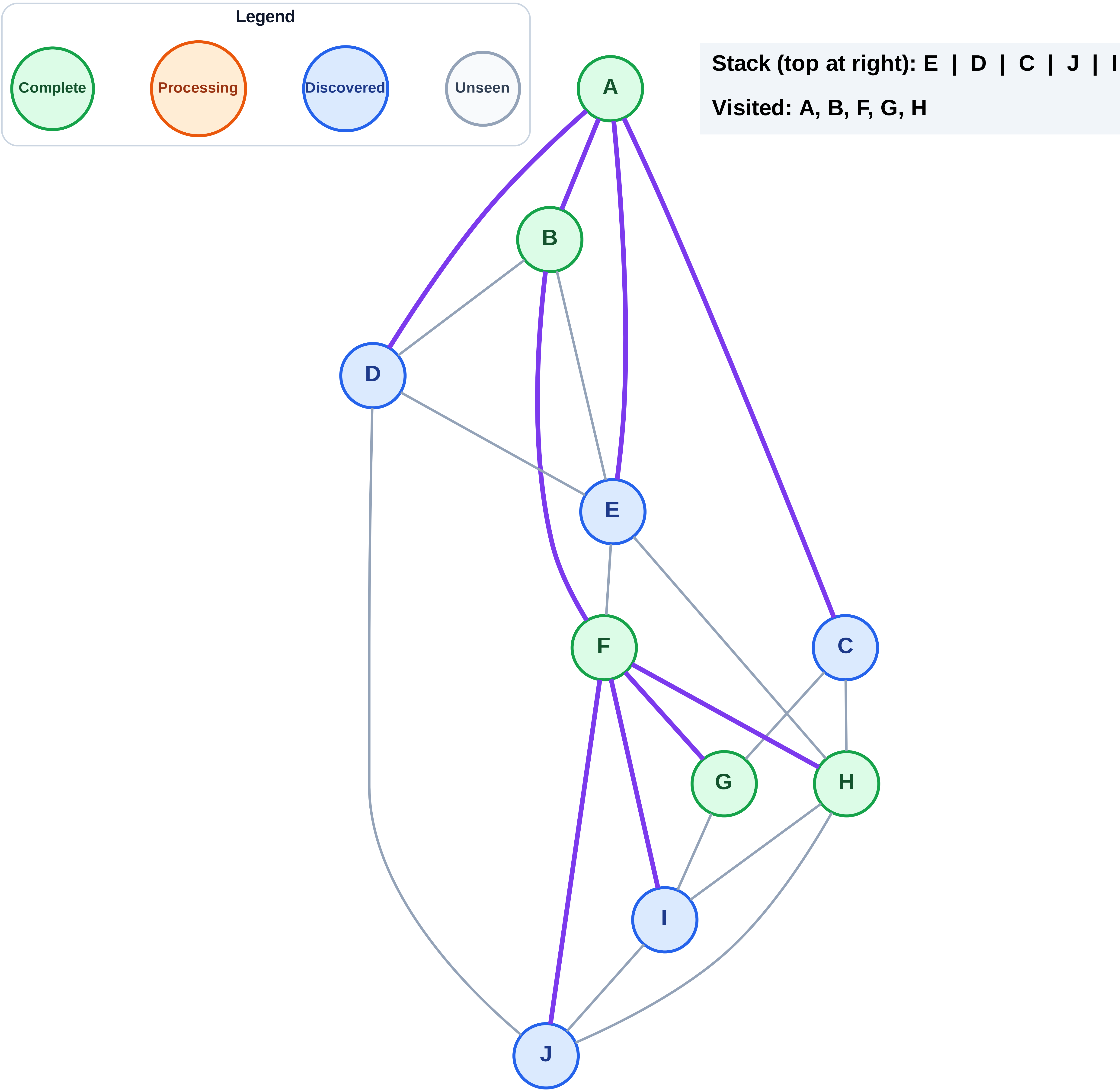
DFS Traversal

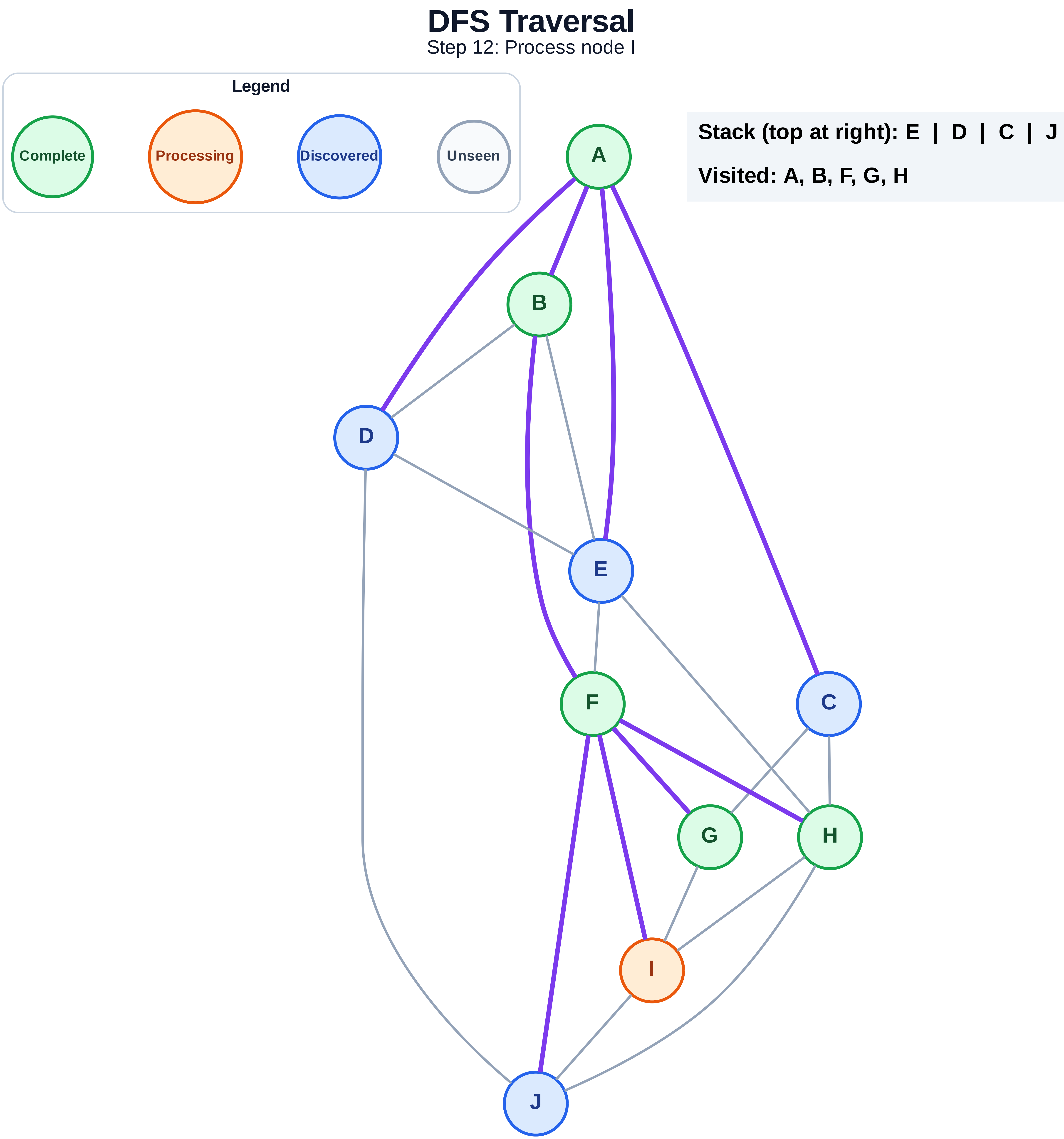
Step 10: Process node H



DFS Traversal

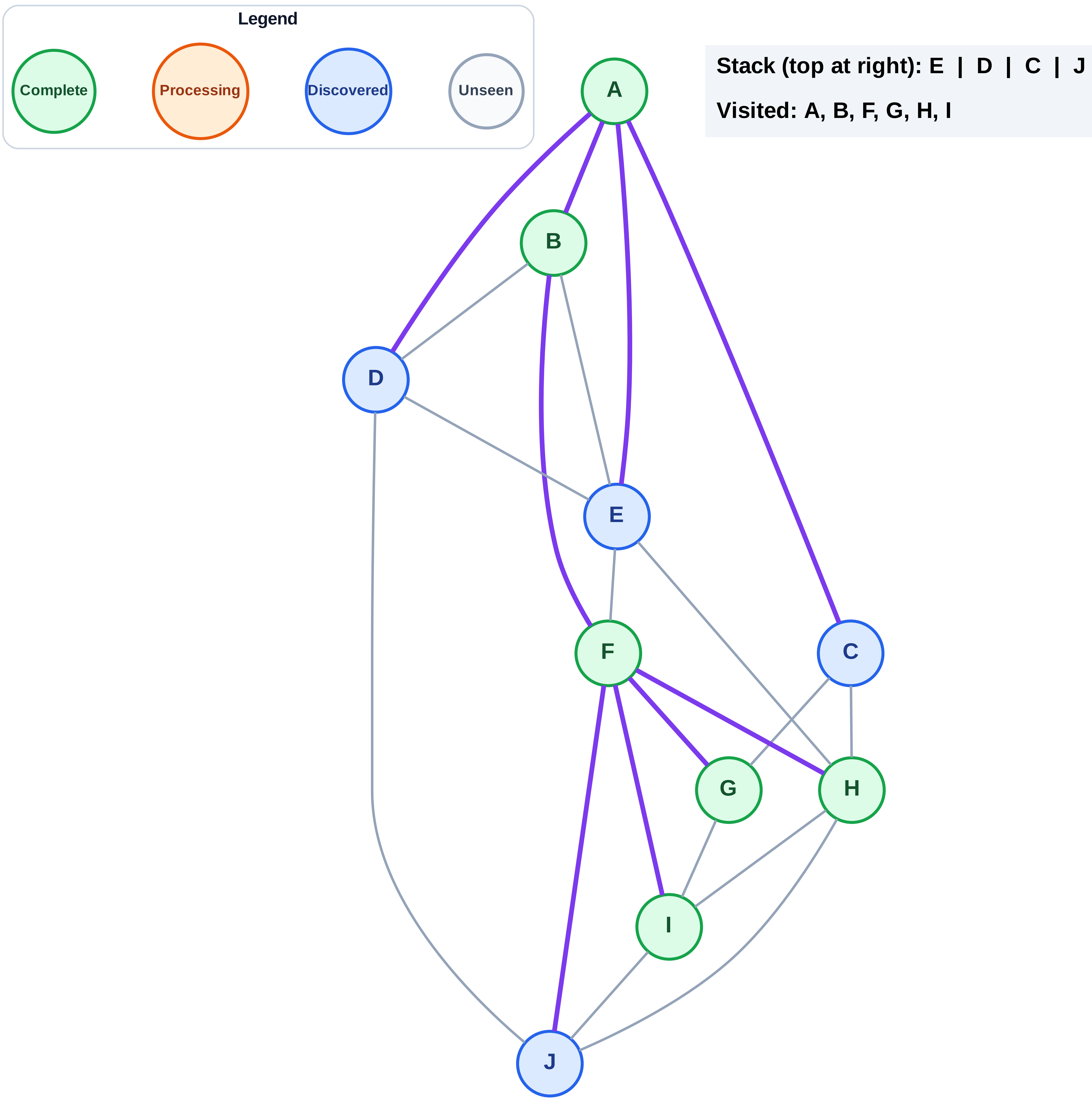
Step 11: No new neighbors from H




DFS Traversal
Step 12: Process node I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): E | D | C | J
Visited: A, B, F, G, H
A
B
C
D
E
F
G
H
I
J

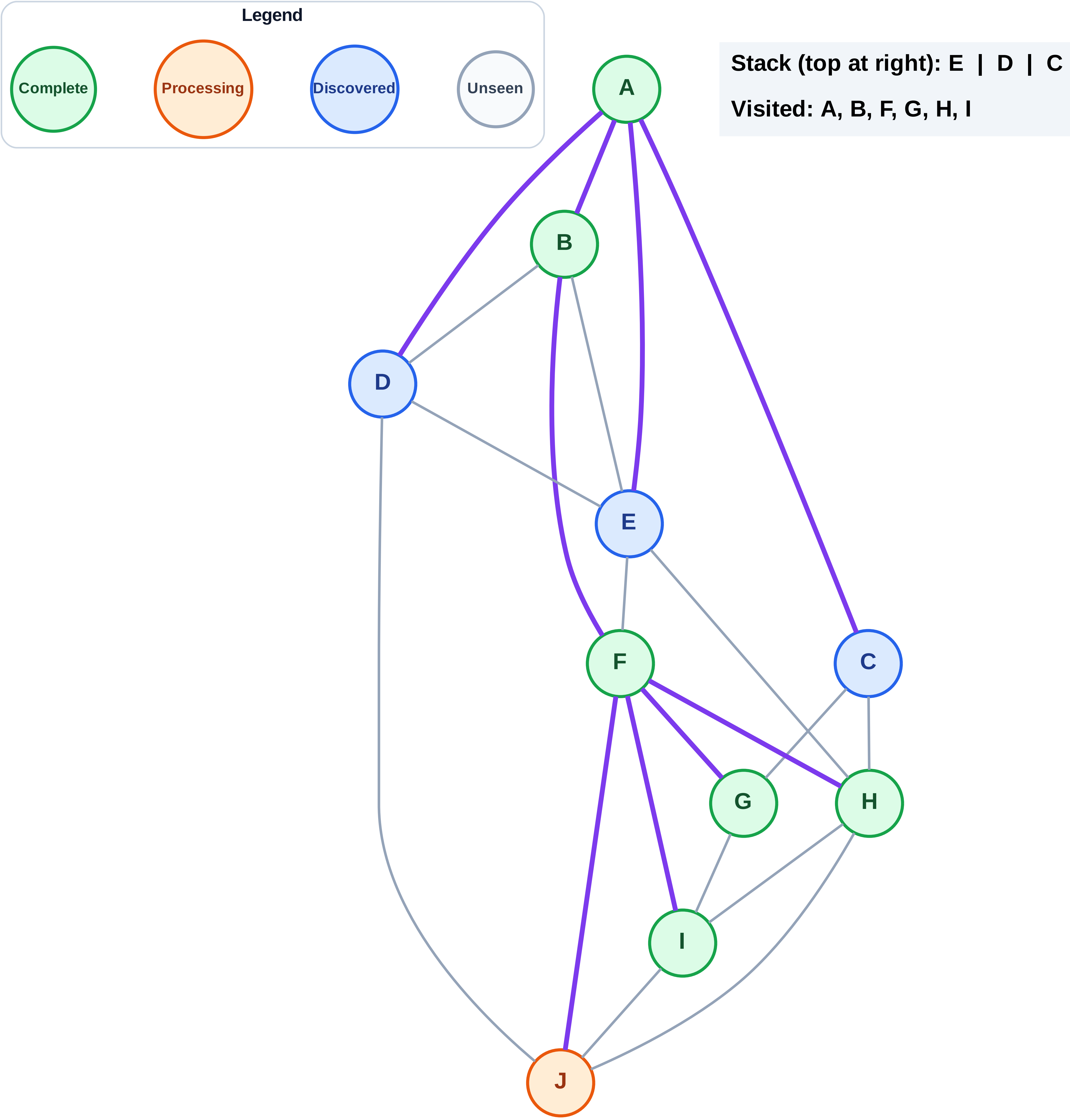
DFS Traversal

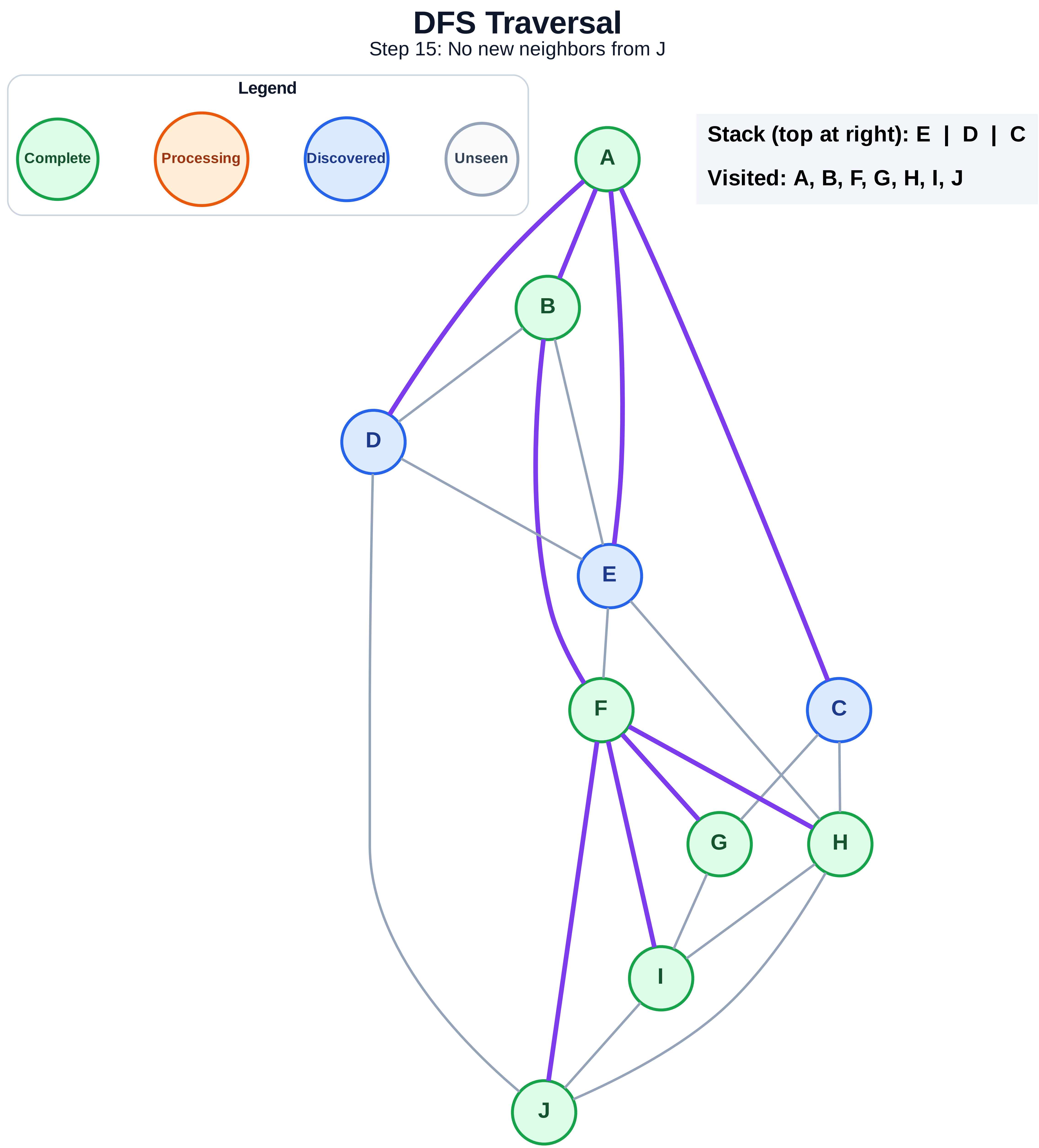
Step 13: No new neighbors from I



DFS Traversal

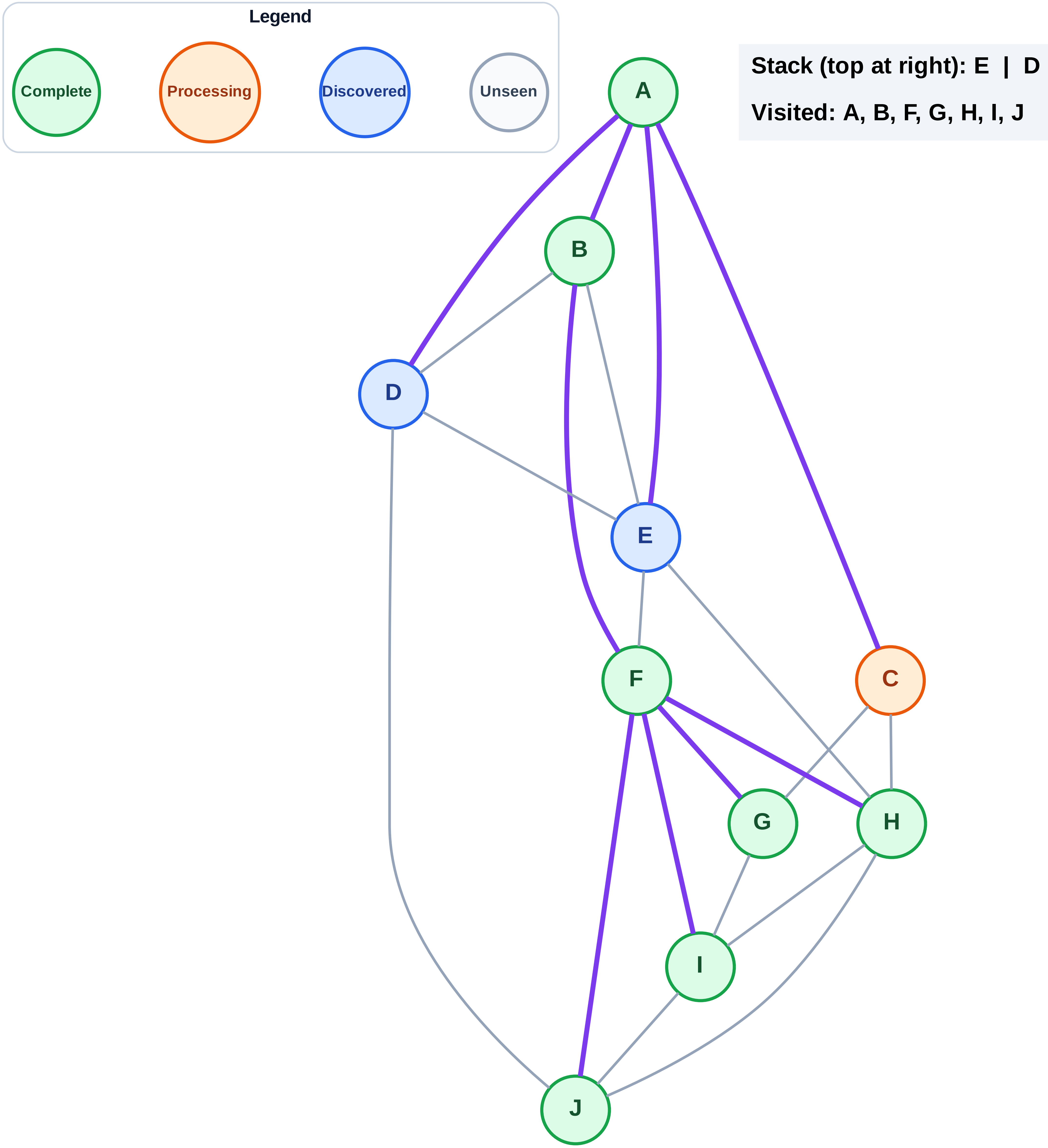
Step 14: Process node J




DFS Traversal
Step 15: No new neighbors from J
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): E | D | C
Visited: A, B, F, G, H, I, J
A
B
C
D
E
F
G
H
I
J

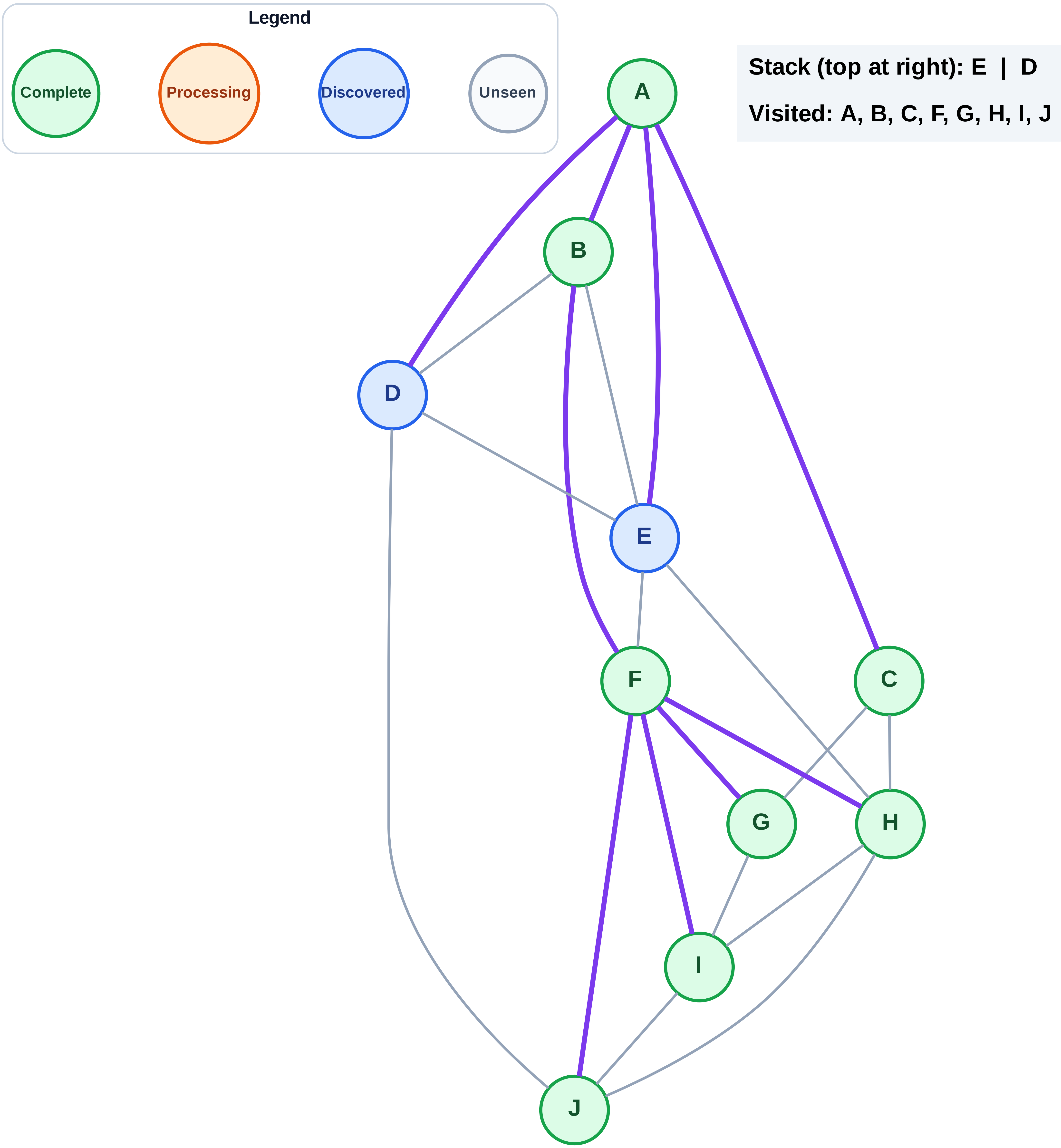
DFS Traversal

Step 16: Process node C



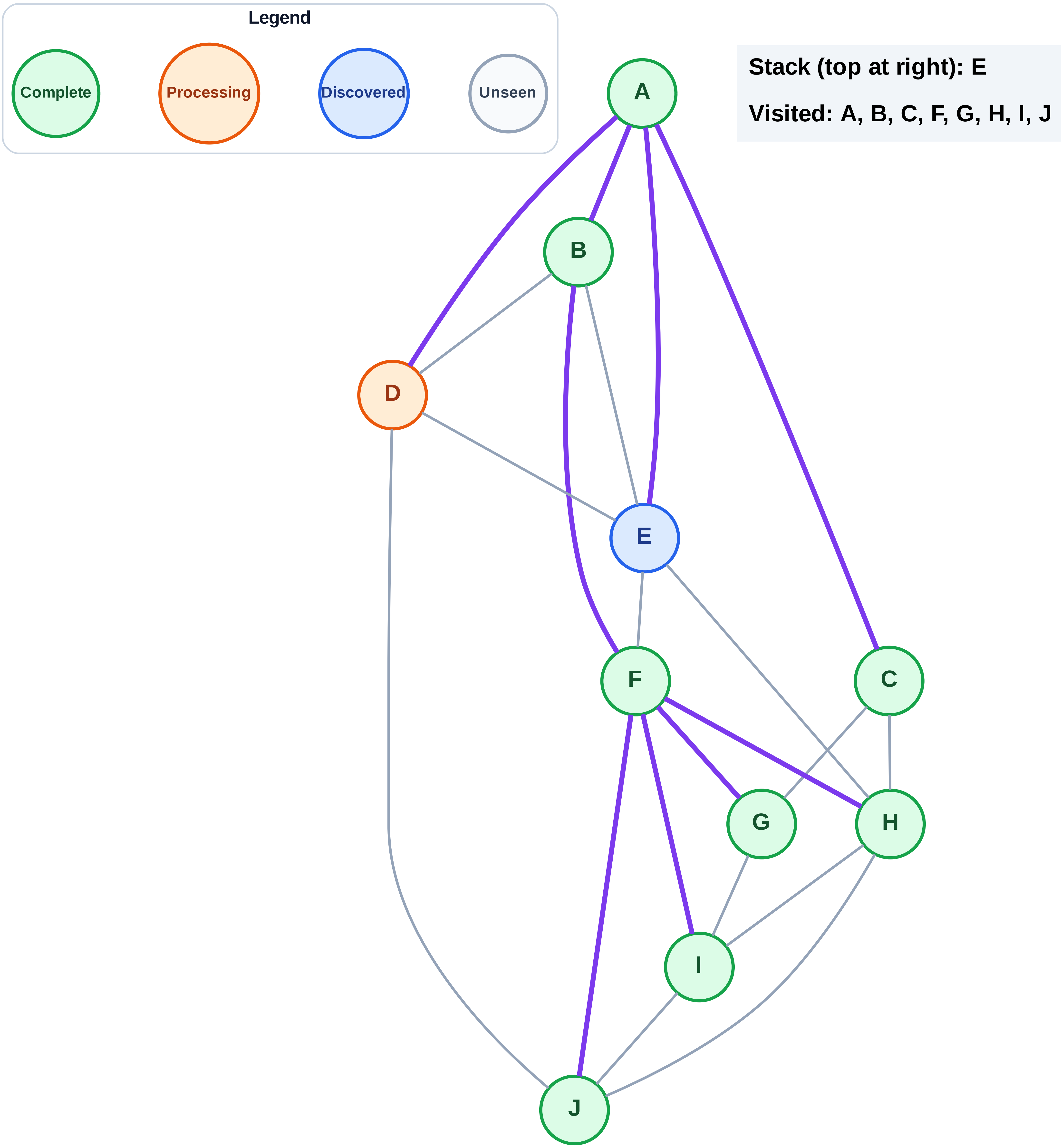
DFS Traversal

Step 17: No new neighbors from C



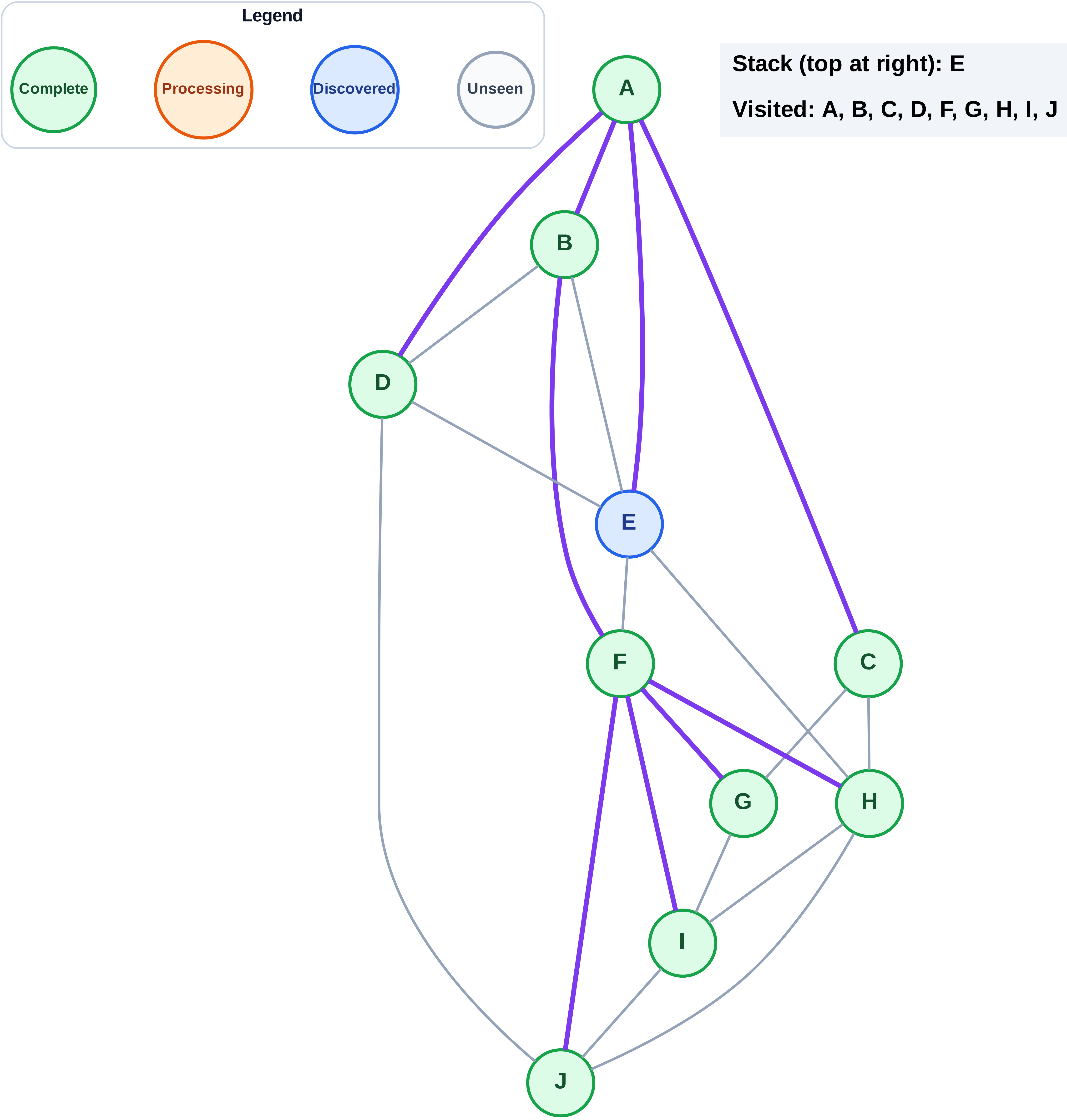
DFS Traversal

Step 18: Process node D



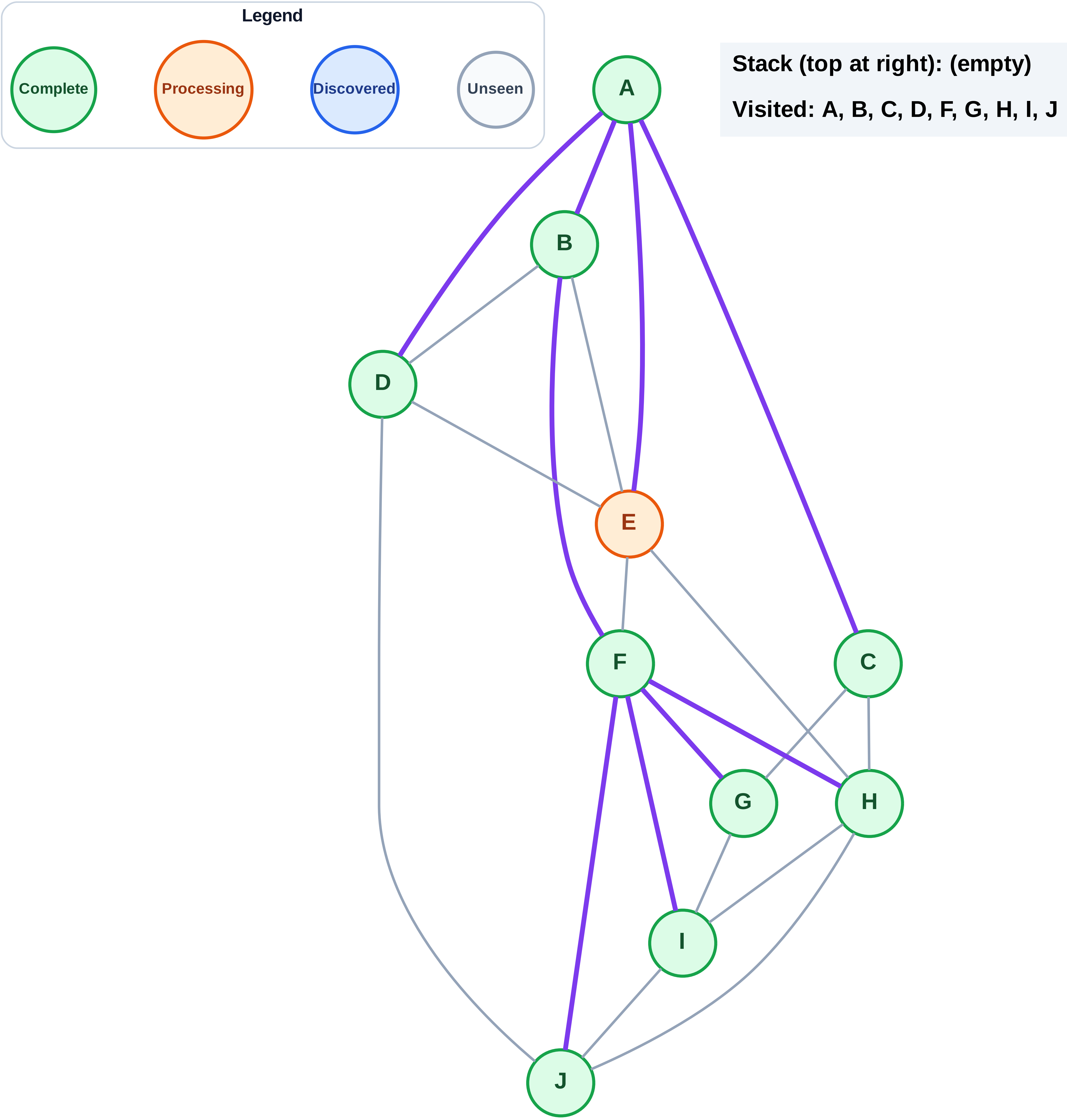
DFS Traversal

Step 19: No new neighbors from D



DFS Traversal

Step 20: Process node E



DFS Traversal

Step 21: No new neighbors from E